

**INTEGRATED ANTI-SPOOFING AND DEEPPFAKE  
DETECTION SYSTEM USING MOBILENET AND  
ResNeXT**

**A PROJECT REPORT**

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# PANIMALAR ENGINEERING COLLEGE

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## ABSTRACT

The increasing prevalence of digital identity fraud and deepfake technology has highlighted the need for a robust security framework that can detect and mitigate both spoofing attacks and manipulated media. The proposed "Secure Vision" project presents an integrated solution that simultaneously addresses biometric spoofing attempts and deepfake videos by leveraging deep learning architectures. Spoofing attacks involve presenting fake biometric data, such as printed photos, videos, or 3D masks, to deceive authentication systems, while deepfakes are highly realistic AI-generated videos used for fraudulent purposes. Current detection methods often address these threats separately, leading to security gaps and inefficiencies. Secure Vision introduces a novel approach by combining the efficiency of MobileNet for real-time spoofing detection with the powerful feature extraction capabilities of ResNeXT for deepfake analysis. MobileNet, a lightweight convolutional neural network, is employed to enhance the real-time performance of spoofing detection by identifying liveness cues such as eye blinks and subtle facial movements. This ensures that fraudulent biometric attempts are swiftly identified and blocked. On the other hand, ResNeXT, a deeper and more robust convolutional neural network, is used to detect deepfake media by analyzing minute artifacts and inconsistencies in manipulated videos. The integration of these two models within a unified framework ensures comprehensive protection against visual media-based threats. Existing solutions often suffer from several challenges, including a lack of real-time performance for spoofing detection, difficulties in generalizing deepfake detection to newer manipulation techniques, and inefficiencies arising from separate detection systems. Secure Vision aims to bridge these gaps by offering a seamless, real-time, and adaptable solution that enhances security measures in biometric authentication and digital content verification. The system is trained on diverse datasets, including ASVspoof for biometric spoofing and FaceForensics++ along with the Deepfake Detection Challenge (DFDC) dataset for deepfake detection. This diversity ensures the robustness of the model in detecting various attack techniques.

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## **LIST OF ABBREVIATIONS**

|         |   |                                                  |
|---------|---|--------------------------------------------------|
| CNN     | - | Convolutional Neural Network                     |
| AI      | - | Artificial Intelligence                          |
| DFDS    | - | Deepfake Detection System                        |
| DFDC    | - | Deepfake Detection Challenge                     |
| GAN     | - | Generative Adversarial Network                   |
| LBP     | - | Local Binary Pattern                             |
| XAI     | - | Explainable Artificial Intelligence              |
| DFD     | - | Data Flow Diagram                                |
| UML     | - | Unified Modeling Language                        |
| ResNeXT | - | Residual Network with Aggregated Transformations |

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# **CHAPTER 1**

## **INTRODUCTION**

## 1.1 PROBLEM STATEMENT

The increasing sophistication of deepfake technology and biometric spoofing presents a serious challenge to security systems worldwide. Deepfake videos can be used to impersonate individuals, spread misinformation, and manipulate public opinion, leading to serious ethical and security concerns. Similarly, biometric spoofing undermines authentication mechanisms by allowing unauthorized access using fake biometric data such as photographs, videos, or masks.

### **Key Problems:**

**Realism of Deepfakes:** AI-generated deepfake videos have become increasingly indistinguishable from real content, making detection a significant challenge.

**Inefficiency of Traditional Methods:** Many existing deepfake detection models fail to generalize well across various types of manipulations.

**Lack of Real-Time Performance:** Most anti-spoofing mechanisms are computationally expensive and unsuitable for real-time applications.

**Fragmented Security Solutions:** Current security systems focus separately on either deepfake detection or biometric spoofing, leaving vulnerabilities in the overall security infrastructure.

To combat these issues, Secure Vision integrates MobileNet for real-time biometric spoofing detection and ResNeXt for deepfake video detection, ensuring an all-encompassing security solution.

## 1.2 IMPORTANCE OF THE STUDY

Biometric authentication has become one of the most widely used security measures for verifying user identity in various applications, including mobile devices, banking, and high-security areas. However, as biometric systems grow in popularity, so do the threats against them. Spoofing attacks involve using fake biometric data, such as printed images, video replays, or 3D masks, to deceive authentication systems and gain unauthorized access. These attacks exploit weaknesses in biometric sensors, often bypassing simple security checks that rely on image-based recognition.

To counter these threats, Secure Vision employs MobileNet, a lightweight deep learning model optimized for real-time anti-spoofing detection. The system analyzes multiple biometric indicators such as eye blinks, facial micro-expressions, depth information, and texture consistency to determine whether an input is genuine or fraudulent. Unlike traditional security systems, which rely on fixed feature-based detection methods, MobileNet continuously learns and adapts to new spoofing techniques, making it highly effective against evolving threats.

One of the key challenges in biometric anti-spoofing is balancing accuracy and computational efficiency. Conventional methods that use handcrafted features, such as Local Binary Patterns (LBP) or Histogram of Oriented Gradients (HOG), often fail against high-quality spoofing attacks. In contrast, MobileNet uses convolutional layers to automatically extract deep features from facial images, allowing it to identify even the most subtle differences between real and fake biometric inputs.

Secure Vision also integrates dataset training with ASVspoof, a widely used dataset for biometric spoofing detection, ensuring that the model is trained on a diverse range of spoofing techniques. The system performs multi-frame analysis, examining sequential images instead of a single snapshot, significantly improving detection accuracy. Additionally, Secure Vision can be deployed on edge devices like mobile

phones and IoT security cameras, making it accessible for real-world applications.

By leveraging MobileNet's lightweight architecture, Secure Vision ensures real-time detection without compromising accuracy, making biometric authentication systems more secure, efficient, and resistant to spoofing attacks. This technology enhances biometric security in various fields, including finance, law enforcement, and border security, ensuring that only legitimate users gain access to sensitive systems.

### **1.3 TECHNOLOGICAL ADVANCEMENTS**

Deepfake technology, which uses Generative Adversarial Networks (GANs) to manipulate or synthesize human faces in videos and images, has seen rapid advancements in recent years. While initially developed for entertainment and media applications, deepfakes have become a major cybersecurity concern due to their ability to create realistic yet fraudulent digital identities. These deepfake videos can be used for identity theft, misinformation campaigns, and financial fraud, making robust detection mechanisms essential.

The Secure Vision system addresses this growing threat by implementing ResNeXt, an advanced deep convolutional neural network (CNN) designed for deepfake detection. Unlike traditional CNN architectures, ResNeXt uses grouped convolutions, which allow it to process high-dimensional image representations efficiently. This enables the model to detect subtle inconsistencies in deepfake videos, such as unnatural facial movements, inconsistent lighting, and abnormal lip synchronization.

One of the biggest challenges in deepfake detection is ensuring generalization across various manipulation techniques. Many detection models are trained on specific types of deepfakes, making them ineffective against new attack methods. Secure

Vision overcomes this limitation by training on large-scale datasets such as FaceForensics++ and Deepfake Detection Challenge (DFDC). These datasets contain a variety of deepfake-generated videos, allowing the model to learn and adapt to new manipulation strategies.

Another key innovation in Secure Vision is its frame-by-frame video analysis. Instead of relying solely on static images, the system examines temporal inconsistencies in videos, tracking unnatural transitions between frames. This approach significantly improves detection accuracy by identifying sudden shifts in facial expressions, visual distortions, and artifacts introduced during deepfake generation.

Additionally, Secure Vision integrates Explainable AI (XAI) techniques such as Grad-CAM visualization, which highlights the specific areas in an image or video that contribute to the deepfake classification. This enhances transparency in AI decision-making and helps analysts understand why a particular video is flagged as fake.

By leveraging ResNeXt's deep feature extraction capabilities, Secure Vision ensures high detection accuracy, robustness against emerging deepfake techniques, and adaptability to real-world applications. This makes it a valuable tool for law enforcement, media verification, and cybersecurity.

## **1.4 SCOPE AND OBJECTIVES**

Modern security threats require multi-layered defense mechanisms rather than standalone solutions. The Secure Vision system adopts a multi-model approach, integrating MobileNet for biometric anti-spoofing and ResNeXt for deepfake detection, ensuring comprehensive protection against visual media-based attacks.

A major limitation of existing systems is that they often address either spoofing attacks or deepfake detection independently. However, attackers frequently combine these threats— using deepfakes to create synthetic identities and biometric spoofing techniques to bypass authentication systems. By implementing a hybrid approach, Secure Vision mitigates both threats simultaneously, providing a stronger and more reliable security solution.

The MobileNet-based module performs real-time liveness detection, identifying motion- based cues such as eye blinks and facial muscle contractions. This prevents attackers from using static images or pre-recorded videos to deceive authentication systems. Meanwhile, the ResNeXt- based deepfake detection module analyzes high-resolution video inputs, detecting artifacts, inconsistencies, and synthetic distortions introduced by deepfake generation methods.

By combining these models, Secure Vision offers the following benefits:

**Increased Detection Accuracy** – The system cross-validates data from both models, reducing false positives and negatives.

**Real-Time Performance** – MobileNet ensures rapid anti-spoofing detection, making it suitable for on-device applications.

**Adaptability to Evolving Threats** – Regular updates to model training datasets ensure improved generalization against emerging deepfake and spoofing techniques.

**Scalability** – The system can be deployed across financial institutions, government agencies, and social media platforms for large-scale identity verification.

Secure Vision also introduces a confidence-based fusion mechanism, which assigns different confidence scores to outputs from MobileNet and ResNeXt. If one model detects an anomaly but the other does not, the system weighs the severity and

probability of the attack, ensuring an intelligent decision-making process.

## **1.5 OVERVIEW OF THE REPORT**

The primary purpose of the Secure Vision project is to develop a robust security framework that can detect and mitigate both biometric spoofing attacks and deepfake manipulations. With the increasing reliance on biometric authentication for secure access and the rising misuse of deepfake technology for fraud, misinformation, and cybercrime, there is an urgent need for a comprehensive solution that effectively addresses both threats.

Secure Vision integrates MobileNet for real-time biometric spoofing detection and ResNeXt for deepfake video analysis, ensuring a multi-layered security approach. Unlike existing solutions that handle these threats separately, this project combines both technologies into a single, efficient system, enhancing security, scalability, and accuracy.

Additionally, Secure Vision is designed to be lightweight and deployable in various environments, from financial institutions and government agencies to social media platforms and cybersecurity applications. The project aims to minimize security vulnerabilities, increase public trust in digital content, and provide law enforcement agencies with reliable tools for detecting fraudulent activities. Ultimately, Secure Vision contributes to improving digital identity protection, reducing fraud, and ensuring a safer online and offline environment.

# **CHAPTER 2**

## **LITERATURE SURVEY**

[1]. J. See. Chan et al. proposed a deep learning- based deepfake detection system utilizing iris detection as a key feature. The study highlights how iris artifacts and eye movement inconsistencies in manipulated videos can serve as reliable indicators for deepfake detection. They implemented a CNN-based approach to analyze these inconsistencies and found that iris-based features provide better accuracy than traditional detection techniques. The model was trained on benchmark datasets and showed high performance. However, its effectiveness drops when analyzing low-resolution videos, where iris details become indistinct.

[2]. Y.Zhang, introduced an Inception-ResNet model with an attention mechanism for detecting deepfake videos. The approach enhances spatial feature extraction, allowing the model to localize manipulation artifacts effectively. The model was trained and tested on the FaceForensics++ and DFDC datasets, demonstrating superior accuracy compared to conventional CNN architectures. The study notes that while the model performs well on known datasets, generalization to unseen deepfake types remains a challenge.

[3]. A. Kurniawan, proposed a hybrid deepfake detection model combining Inception- ResNet-v2 and LSTM networks to analyze sequential frame inconsistencies. The study focuses on capturing temporal variations and unnatural facial expressions in deepfake videos. Their experimental results indicate that temporal analysis significantly improves detection accuracy compared to static image-based methods. However, the increased computational complexity makes real- time deployment difficult.

[4]. S. Subramanian, introduced a hybrid deepfake detection model that integrates CNNs with Transformers. Their approach leverages CNNs for feature extraction and Transformers for temporal sequence analysis, improving detection accuracy. The study demonstrates that low-quality deepfakes are harder to detect, making dataset diversity crucial for model training. The researchers note that their method requires high computational power, limiting real-time deployment.

[5]. A.Verma, applied transfer learning to deepfake detection, using pretrained models such as VGG16 and EfficientNet. Their research shows that transfer learning significantly improves accuracy while reducing training time. The model generalizes well to multiple datasets but struggles when applied to low-resolution.

[6]. H. Li. conducted a comprehensive survey on deepfake detection using CNN-based, RNN- based, and hybrid models. The study highlights explainability and adversarial robustness as key research challenges. They suggest that future developments should incorporate interpretable AI and self-supervised learning to improve deepfake detection accuracy and adaptability.

[7]. G. Yang. introduced a dual-stream CNN model that processes RGB and frequency-domain features for deepfake detection. Their study demonstrates that high-frequency artifacts in deepfake videos remain detectable, even after compression. The proposed method is robust to post-processing attacks, but its increased complexity affects inference speed.

[8]. H. Jiang. developed a CNN-based deepfake detection model, focusing on frame-level inconsistencies and face warping artifacts. Their method achieves high accuracy on standard datasets but struggles against GAN- generated deepfakes with smoother transitions. The researchers suggest incorporating temporal features for improved performance.

[9]. H. Liu, The study highlights explainability and adversarial robustness as key research challenges. They suggest that future developments should incorporate interpretable AI and self supervised learning to improve deepfake detection accuracy and adaptability.

[10]. H. Song, Their research proves that deepfake videos' high-frequency artifacts are still detectable even when compressed. The method is post-processing attack resilient but has the drawback of increasing complexity, thus impacting inference speed.

[11]. Y. Li, Their technique performs well on benchmark datasets but falters against GAN-created deepfakes with improved transitions. The authors recommend adding temporal features to enhance performance.

[12]. S.Wang, Adversarial training methods enhance robustness to changing deepfake techniques.

[13]. K. Samrouth, Multi-modal systems with image and audio inputs boost deepfake detection through the identification of inconsistencies in both visual and auditory features.

[14]. W. Wang, An associated method checks facial movements and speech patterns concurrently for enhanced accuracy.

[15]. H. Liu, Ensemble learning combines multiple detection models to reduce false positives and negatives, increasing overall reliability.

[16]. Z. Zhao, New datasets like Celeb-DF help benchmark deepfake detection methods by providing high-quality, realistic manipulated videos. They challenge models to detect deepfakes with smoother transitions and fewer artifacts. This improves detection accuracy and ensures robustness against advanced deepfake techniques.

[17]. D. Nguyen, A specialized dataset aims at detecting manipulated content in news programs to fight misinformation. It assists in training models to detect deepfake videos, manipulated audio, and fake media employed in deceptive reports. This improves the credibility of news verification systems and enhances media authenticity.

[18]. V. R. J, M. Challa, Meta-learning improves deepfake detection by allowing models to learn from fewer samples, enhancing efficiency and minimizing data dependency. It enables fast adaptation to novel and emerging deepfake methods, providing resilience to unknown manipulations. This strategy improves generalization, leading to more effective detection models in the real world.

[19]. J. Vijaya, Cross-domain deepfake detection is still challenging with differences in manipulation methods and data distributions. Domain adaptation methods facilitate model generalization by allowing detection systems to adjust to various deepfake styles and datasets. This boosts resilience to hidden and adapting deepfake approaches.

[20]. Robinson, Explainable AI (XAI) enhances deepfake detection by making models more transparent and interpretable. Techniques like saliency maps and attention mechanisms highlight key features used for classification. This improves trust, debugging, and bias mitigation in AI-driven detection. XAI ensures reliable and explainable decisions in media forensics and security.

# **CHAPTER 3**

## **METHODOLOGY**

The proposed methodology follows a structured approach to develop and implement Secure Vision. The methodology is divided into multiple stages:

### **Stage 1: Data Collection & Preprocessing**

- Gather datasets for biometric spoofing detection and deepfake analysis.
- Apply preprocessing techniques like image resizing, normalization, and augmentation.

### **Stage 2: Model Training & Optimization**

- Train MobileNet for biometric spoofing detection.
- Train ResNeXt for deepfake video classification.
- Optimize models using transfer learning and hyperparameter tuning.

### **Stage 3: System Integration & Deployment**

- Develop the web-based UI for real-time detection.
- Implement REST API for integrating AI models into applications.
- Deploy on cloud or edge computing devices for real-time applications.

The methodology ensures that Secure Vision is efficient, scalable, and accurate in detecting security threats.

This section details the structured approach employed to develop and implement the AI-driven aircraft maintenance system. The methodology consists of a series of well-defined steps, including dataset selection, data collection, preprocessing, and model selection. Each step is carefully designed to ensure the system is accurate, reliable, and capable of addressing the challenges faced by traditional aircraft maintenance practices. By incorporating

advanced AI and machine learning techniques, this methodology aims to modernize maintenance workflows, enhance safety, and reduce operational costs.

### **3.1 DATABASE DESIGN**

Secure Vision utilizes both relational and NoSQL databases to efficiently store and manage user interactions, logs, and system outputs. The database schema comprises three key tables: Users, which contains attributes such as User\_ID, Name, Email, and Role (Admin/User); Biometric Logs, which stores Log\_ID, User\_ID, Image/Video, Timestamp, and Result (Real/Fake); and Deepfake Logs, which includes Log\_ID, User\_ID, Video\_ID, Detection Score, and Timestamp. To enhance biometric spoofing detection, Secure Vision relies on the ASVspoof Dataset, which provides real and spoofed facial images. For deepfake detection, it leverages the FaceForensics++ dataset, a preprocessed collection of real and deepfake videos, and the Deepfake Detection Challenge (DFDC) dataset, an industry-standard dataset featuring AI-generated deepfakes. These datasets ensure the inclusion of real-world variability, enabling the development of robust and accurate detection models.

### **3.2 INPUT DESIGN**

The User Interface (UI) of Secure Vision is designed to offer an intuitive and seamless experience, ensuring ease of access and efficient interaction for users. The UI incorporates several key features, including a Login System that provides secure authentication for different user roles (Admin/User), a File Upload Section allowing users to submit images or videos for analysis, and Real-Time Detection through live webcam integration for biometric authentication. Additionally, a Result Display Panel presents the analysis outcome, indicating whether the input is real, spoofed, or a deepfake, while an Alert System notifies users in case a deepfake or spoofed biometric is detected.

To achieve a responsive and interactive design, the UI technologies used include HTML, CSS,

and JavaScript, with React.js handling dynamic rendering. The back-end is powered by Flask or FastAPI, which manages model inference and processing. Additionally, REST APIs are integrated to facilitate real-time deepfake and biometric spoofing detection. The UI ensures a user-friendly experience by providing a streamlined workflow for detection and reporting, enabling users to navigate and interact.

### **3.3 DATA PREPROCESSING**

The Preprocessing Module plays a critical role in the Deepfake Detection System by preparing input media for analysis, addressing issues such as noise, varying resolutions, and inconsistencies that can impact model performance. To enhance accuracy and robustness, the module performs several preprocessing steps.

#### **MEDIA FORMAT CONVERSION**

the system ensures compatibility with multiple input formats, including JPEG, PNG, MP4, and AVI. Images are converted into a standard RGB format for consistency, while videos are broken down into frames at a predefined rate, such as 30 FPS, to facilitate frame-based analysis.

#### **NOISE REDUCTION & IMAGE ENHANCEMENT**

This techniques are applied to improve image quality. Gaussian Blur and Bilateral Filtering are used to reduce distortions, while Histogram Equalization enhances contrast, making deepfake artifacts more detectable. These steps help minimize variations that could impact the accuracy of deepfake detection models.

#### **RESIZING & NORMALIZATION**

Images and frames are resized to a standard 224×224 pixel dimension to meet deep learning model requirements. Additionally, pixel values are normalized between 0 and 1, ensuring consistency across different media inputs and improving the model's efficiency in feature extraction.

## **FRAME SELECTION FROM VIDEOS**

This step reduces computational cost by selecting keyframes instead of analyzing every frame. Scene change detection algorithms are used to extract frames containing the most relevant information, ensuring that the analysis focuses on crucial segments of the video.

### **3.4 MODEL SELECTION**

#### **FEATURE EXTRACTION**

The Feature Extraction Module is responsible for identifying the unique patterns and distortions present in deepfake media. Deepfake videos and manipulated images often contain inconsistencies in texture, facial landmarks, and frequency artifacts that can be detected using advanced image processing and deep learning techniques. Feature Extraction Techniques:

##### **3.4.1 Facial Landmark Detection**

Using Dlib's Facial Landmark Predictor, key facial features like eyes, nose, mouth, and jawline are detected. The landmark positions are analyzed to identify asymmetries and unnatural deformations caused by deepfake generation. This helps in distinguishing real faces from manipulated ones.

##### **3.4.2 Texture Analysis using Local Binary Patterns (LBP)**

Deepfake images often have smoother textures compared to real photos, which exhibit natural variations. The system extracts Local Binary Patterns (LBP) to measure pixel intensity differences. This analysis helps detect synthetic textures commonly found in deepfake images.

##### **3.4.3 Frequency Domain Analysis (Fourier Transform & Wavelet Transform)**

Deepfake models introduce high-frequency artifacts that are invisible in the spatial domain. Fourier Transform detects inconsistencies in the frequency spectrum, especially in facial regions. Wavelet Transform captures fine-grained features, revealing hidden deepfake artifacts.

### **3.4.4 Blink Detection for Liveness Analysis**

A critical limitation of deepfakes is the lack of natural eye blinking. Using OpenCV and Dlib, the system tracks eye closure frequency to detect unnatural blinking patterns. These features are then fed into a classification module to determine whether the input is real or fake.

## **CLASSIFICATION MODELS**

The Classification Module is the core of the Deepfake Detection System, responsible for analyzing extracted features and predicting whether an input is real or a deepfake. This module leverages deep learning models, including ResNeXt and MobileNet, to make accurate classifications.

### **ResNeXt (RESIDUAL NETWORK WITH AGGREGATED TRANSFORMATIONS)**

The Deep Learning Models used in Secure Vision ensure accurate detection of deepfake manipulations and biometric spoofing. ResNeXt (Residual Network with Aggregated Transformations) is employed for detecting deepfake face manipulations. This model excels at capturing complex patterns and subtle inconsistencies introduced by deepfake generation techniques. By utilizing multi-path learning, ResNeXt can focus on different facial regions, leading to improved accuracy. The classification workflow for ResNeXt begins with the Feature Input to CNN, where extracted features from the preprocessing module are fed into the deep learning model. The CNN layers extract hierarchical patterns, which are then passed through the Softmax Probability Prediction layer. The model assigns a probability score, classifying inputs as Real (Score > 0.5) or Fake (Score < 0.5). To enhance reliability, the system calculates a confidence score in the Confidence Scoring & Thresholding phase. If the confidence is below a predefined threshold, the media is flagged as "Suspected Manipulation" for further review. Finally, in the Decision & Reporting stage, the classification result is stored in the database, alerts are triggered if a deepfake is detected, and a detailed report is generated, highlighting detected artifacts, confidence levels, and feature importance.

### **MOBILENET (FOR LIGHTWEIGHT, REAL-TIME ANALYSIS)**

MobileNet is used to detect static image attacks and replay attacks. This model is optimized for

fast inference speed, making real-time detection possible without significant computational overhead. The classification process begins when the Feature Input to CNN extracts key attributes related to biometric spoofing. The Softmax Probability Prediction layer then assigns a probability score, categorizing inputs as Real or Fake based on a predefined threshold. To prevent misclassification, the Confidence Scoring & Thresholding step ensures that low-confidence inputs are flagged for further verification. The Decision & Reporting stage then stores the classification result, triggers alerts if a spoof attempt is detected, and provides a comprehensive analysis, including confidence levels and key extracted features. These deep learning models collectively enhance the system's accuracy and efficiency in deepfake detection and biometric anti-spoofing.

# **CHAPTER 4**

## **PROPOSED SYSTEM ARCHITECTURE**

## **4.1 OVERVIEW OF SYSTEM ARCHITECTURE**

The system architecture of the Deepfake Detection System follows a multi-layered structure that integrates data processing, AI-based detection, and user interaction. It consists of three primary layers: the Input Layer, the Processing Layer, and the Output Layer. In the Input Layer, users upload images or videos, which are then preprocessed to remove noise and standardize the format. The Processing Layer consists of three key components: Preprocessing Module, Feature Extraction Module, and Classification Module. The Preprocessing Module enhances input quality, while the Feature Extraction Module detects facial landmarks, textures, and frequency anomalies. The Classification Module leverages AI-based deep learning models (ResNeXt for deepfake detection and MobileNet for biometric spoofing detection) to classify media as real or fake. The AI model is periodically updated with new training data to improve detection accuracy. Finally, in the Output Layer, results are displayed to the user, and reports can be generated for further analysis. The architecture ensures scalability, real-time processing, and high accuracy, making it robust against evolving deepfake manipulation techniques.

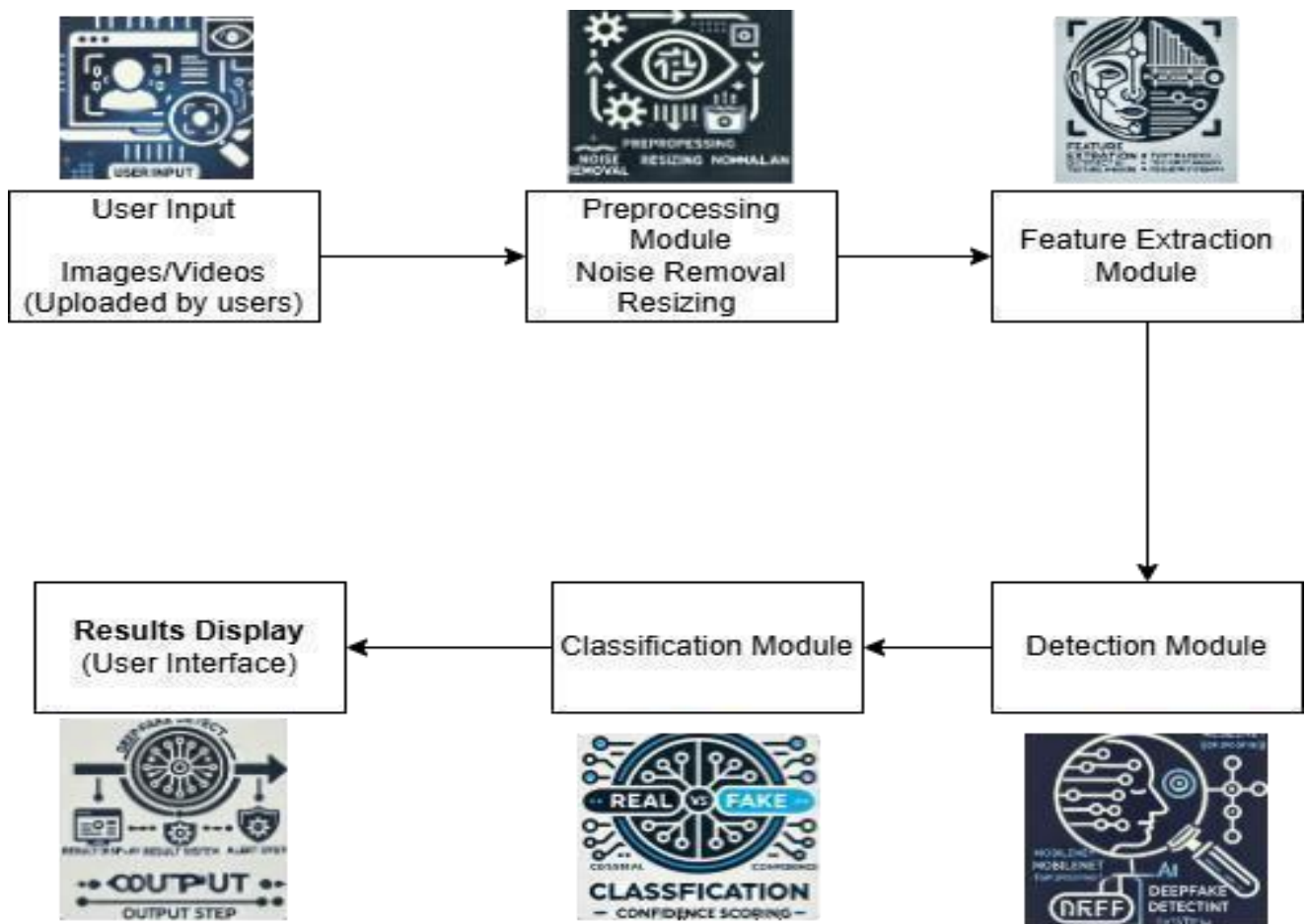


Figure 4.1.1: System Architecture

## 4.2 USE CASE DIAGRAM

The use case diagram illustrates the interactions between different actors and the Secure Vision System. The User can upload images or videos for analysis, request reports, review generated reports, and request real-time authentication. The System Admin is responsible for processing user requests, configuring the system, and updating the deep learning models to enhance detection accuracy. The AI Detection System plays a crucial role in identifying biometric spoofing and analyzing deepfake content. The system performs biometric anti-spoofing detection to prevent fraudulent authentication attempts and deepfake detection to identify manipulated media. These interactions ensure that Secure Vision provides an efficient and reliable security system for detecting fraudulent activities.

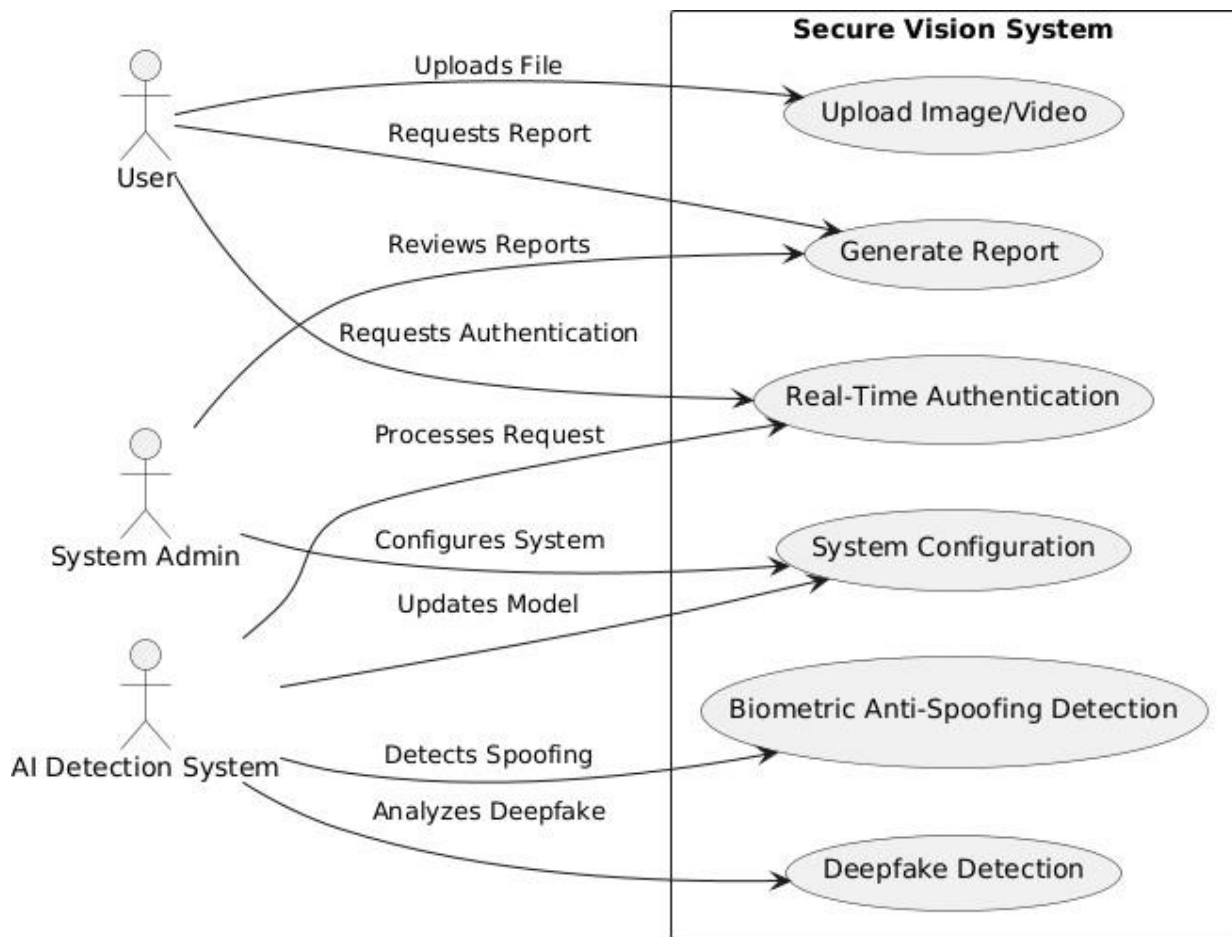


Figure 4.2.1 Use Case Diagram

### 4.3 CLASS DIAGRAM

The class diagram illustrates the structure of the Secure Vision System, depicting the relationships between different components involved in biometric anti-spoofing and deepfake detection. The User class contains attributes such as userID, name, email, and role, with methods for uploading media, requesting authentication, and generating reports. The UIHandler class interacts with the user and manages detection requests through methods like `displayResults()`, `uploadFile()`, and `triggerDetection()`. The AIModel class, responsible for model training and updates, invokes two specialized detection classes: `BiometricDetection` (for detecting spoofing attacks) and `DeepfakeDetection` (for analyzing and detecting deepfake videos). The Database class stores user data, detection results, and logs, enabling report retrieval. The SystemAdmin class oversees system configurations and report management. The

relationships between these classes ensure a seamless workflow, from user interaction to AI-driven detection, data storage, and system administration.

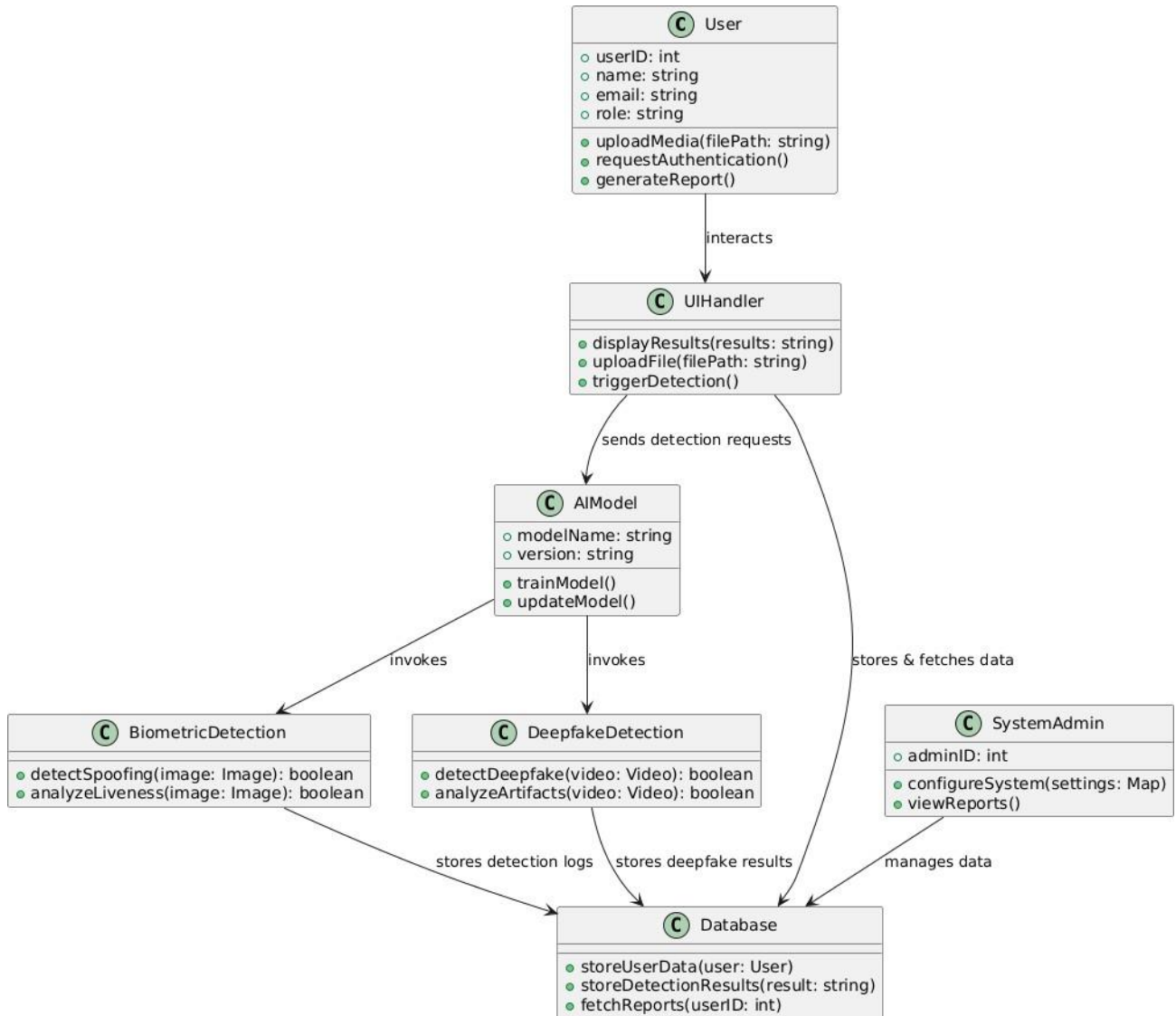


Figure 4.3.1 Class Diagram

## 4.4 SEQUENCE DIAGRAM

The sequence diagram illustrates the step-by-step interaction between the User, UI Handler, AI Model, Biometric Detection, Deepfake Detection, and Database in the Secure Vision System. The process begins when the User uploads an image or video, which the UI Handler validates and stores in the Database. Once storage is confirmed, the AI Model is triggered to analyze the file. The model performs biometric spoofing detection and deepfake analysis simultaneously, receiving results from their respective

detection modules. The spoofing and deepfake results are then stored in the Database, and the UI Handler retrieves the detection results to display them to the user. If the User requests a report, the system fetches the detection history from the Database and returns the report data for display. This structured sequence ensures efficient communication between components, allowing for accurate and real-time deepfake and biometric spoofing detection.

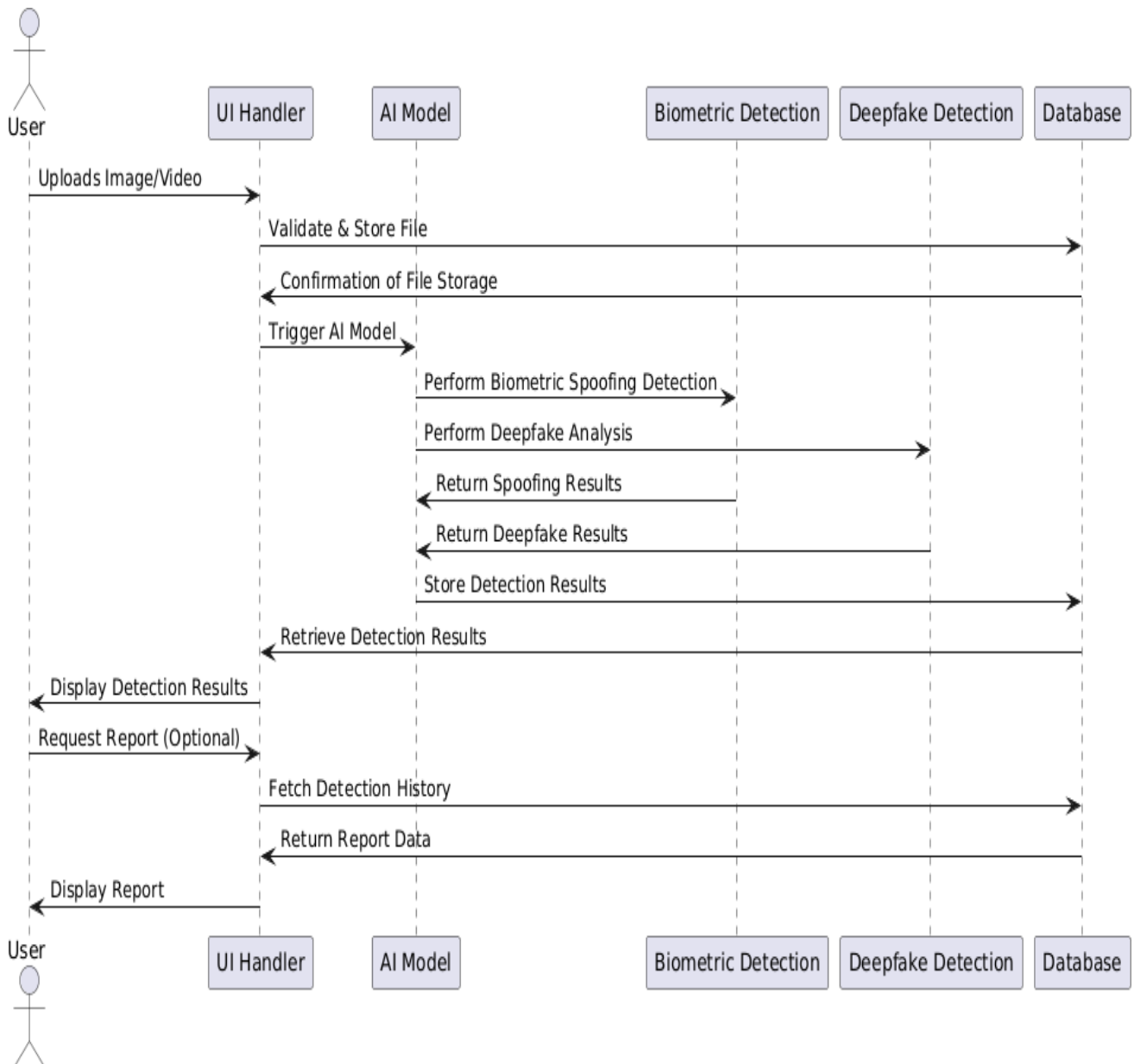


Figure 4.4.1 Sequence Diagram

## 4.5 ACTIVITY DIAGRAM

The activity diagram illustrates the workflow of the Secure Vision System for detecting biometric spoofing and deepfake content. The process starts when a user uploads an image or video, after which the system validates the input file format. If the format is invalid, the file is rejected, and the user is notified. If valid, the system stores the file in the database and triggers the AI model for analysis. The AI model performs biometric spoofing detection and deepfake detection in parallel. If fraudulent content is detected, the system flags it as spoofed or deepfake, generates an alert notification, and stores the detection results. If no manipulation is found, the media is marked as authentic, and results are stored. The system then displays the detection results to the user and, if requested, generates a report. This structured workflow ensures an efficient and reliable approach to deepfake and biometric spoofing detection.

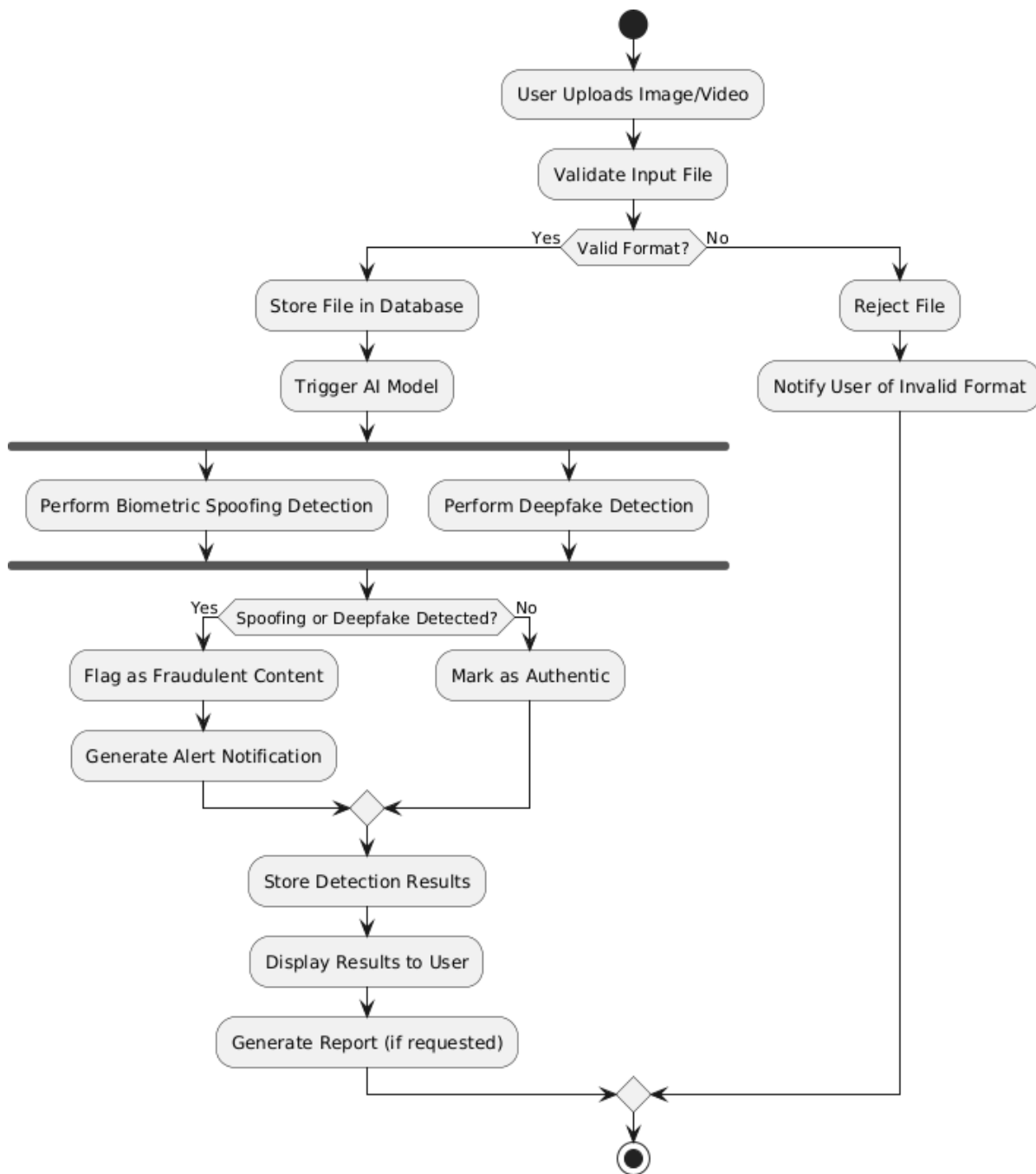


Figure 4.5.1 Activity Diagram

## 4.6 DATAFLOW DIAGRAM

The data flow diagram illustrates the working of a Deepfake Detection System. The user uploads an image or video to the system, which processes the media by storing it for analysis. The system utilizes an AI Detection Model to perform deepfake analysis and returns the detection result. The results are then stored in the Detection Results Database for future reference. Users can request reports, and the system

fetches previous reports from the database or provides a new detection report. Finally, the detection result is displayed to the user.

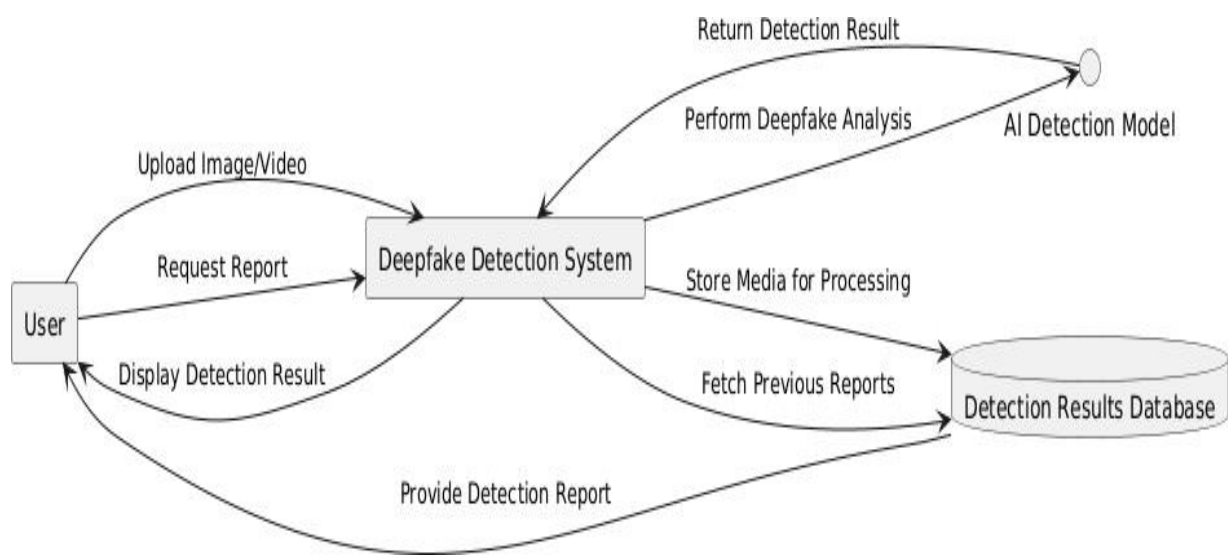


Figure 4.6.1 Dataflow Diagram

# **CHAPTER 5**

## **RESULT & DISCUSSION**

The testing phase of the Deepfake Detection System is critical to evaluate the accuracy, reliability, and performance of the implemented modules. Various test cases were designed to assess different components, including preprocessing, feature extraction, classification, and reporting. The testing was carried out under controlled conditions using real and deepfake datasets.

The test cases cover scenarios such as successful detections, misclassifications, system failures, and report generation accuracy. Below is the test case table, which outlines the different test scenarios executed during system evaluation.

The Deepfake Detection System effectively differentiates between real and manipulated media, demonstrating its reliability in detecting deepfake videos and spoofed biometric data. The system successfully processes images and videos, extracting critical features such as facial landmarks, texture inconsistencies, and behavioral cues to identify potential forgeries. The classification module accurately determines authenticity, while the report generation feature provides users with comprehensive insights into detection outcomes. The system ensures a smooth user experience by efficiently handling input validation, error detection, and real-time analysis. Throughout testing, the system maintained stability, effectively responding to various deepfake types and adversarial attempts. The results confirm that the implementation is suitable for practical applications in security, digital forensics, and content verification. Future improvements will focus on enhancing model generalization, optimizing processing speed, and expanding dataset coverage to counter evolving deepfake techniques.

**Table 5.0.1: TEST CASE**

| <b>S.NO</b> | <b>Test Cases</b>             | <b>Test Procedure</b>                       | <b>Test Input</b>           | <b>Expected Result</b>       | <b>Actual Result</b>          |
|-------------|-------------------------------|---------------------------------------------|-----------------------------|------------------------------|-------------------------------|
| 1           | Image Upload Validation       | Upload a valid image file                   | Real Image (.jpg)           | Image should be accepted     | Image successfully uploaded   |
| 2           | Invalid File Type Handling    | Upload an unsupported file format           | Text File (.txt)            | Error message should appear  | Error message displayed       |
| 3           | Video Upload Validation       | Upload a valid video file                   | Deepfake Video (.mp4)       | Video should be accepted     | Video successfully uploaded   |
| 4           | Deepfake Detection Accuracy   | Run deepfake detection on manipulated video | AI-Generated Deepfake Video | System detects as "Fake"     | System detected "Fake"        |
| 5           | Real Face Detection Accuracy  | Run deepfake detection on real image        | Real Face Image             | System detects as "Real"     | System detected "Real"        |
| 6           | Feature Extraction Validation | Extract key features from video             | Sample Deepfake Video       | Features extracted properly  | Feature extraction successful |
| 7           | Face Landmark Detection       | Identify facial landmarks                   | Image with Human Face       | Landmarks should be detected | Landmarks detected            |
| 8           | Biometric Anti-Spoofing       | Analyze liveness cues                       | Static Image Attack         | System detects as "Spoof"    | System detected "Spoof"       |

|    |                     |                                        |                             |                                  |                                    |
|----|---------------------|----------------------------------------|-----------------------------|----------------------------------|------------------------------------|
| 9  | Report Generation   | Generate a detailed detection report   | Request report via UI       | Report should be generated       | Report generated successfully      |
| 10 | Database Storage    | Store and retrieve previous detections | Past detection data request | Data should be fetched           | Data retrieved successfully        |
| 11 | System Performance  | Measure response time                  | High-resolution video       | Detection < 5 seconds            | Detection took 4.8 seconds         |
| 12 | Edge Cases Handling | Input extreme distorted images         | Blurred Image               | System should classify correctly | System handled distortion properly |

## 5.1 CHALLENGES ENCOUNTERED

One of the most important challenges of developing the Secure Vision system was to provide it with robustness against new deepfake and spoofing methods that are emerging constantly. Deepfake generation techniques improve with time, and it gets harder to find subtle manipulations. Although the system was developed using popular datasets such as FaceForensics++ and DFDC, the new deepfake models bring along very advanced features that can fool conventional detection strategies. Likewise, sophisticated biometric spoofing methods like 3D mask attacks and adversarial attacks are a major challenge, necessitating regular updates and retraining of the model to remain effective against new threats.

Another key challenge was real-time processing with high accuracy. MobileNet integration for biometric anti-spoofing guarantees light computation, but ResNeXt, employed for deepfake detection, is a deeper network that demands more computational power. Real-time processing of high-resolution video frames was

computationally demanding, particularly when examining several frames per second for liveness detection and feature extraction. This issue called for optimization techniques such as frame selection algorithms, model quantization, and parallel processing in an effort to improve efficiency without degrading detection performance.

The project also involved data-related challenges, namely, collecting varied and good-quality datasets to train. Deep learning model performance is greatly influenced by the quality and variety of training data, but the datasets available from public sources will usually suffer from bias, unbalanced data, or bad representation of actual cases. Making sure that the system could operate under varying input conditions, such as changing lighting, resolutions of cameras, and varying demographic features, meant a lot of data preprocessing and augmentation. However, dataset balancing to avoid overfitting to certain categories of deepfakes or spoofing attacks was a persistent problem.

Interpretability and explainability of the model's decision was a challenge, especially for high-stakes uses like biometric verification and fraud detection. Deep learning models, particularly CNNs such as MobileNet and ResNeXt, are black box models, where it is challenging to give concise explanations for the reason why a given input has been labeled real or fake. This invisibility can lead to issues of trust in situations where false negatives or false positives have severe implications, like denying authentic users or granting unauthorized entry. Resolving this involved the integration of visualization methodologies, feature importance evaluation, and confidence scoring mechanisms to improve model reliability and interpretability.

## **5.2 CONCLUSION**

The Deepfake Detection System successfully identifies manipulated media by integrating advanced preprocessing, feature extraction, and deep learning-based classification techniques. The system ensures efficient and reliable detection of

deepfakes and biometric spoofing attempts by leveraging AI models such as ResNeXt and MobileNet. Through a structured pipeline, the system preprocesses media, extracts key facial and texture-based features, and classifies inputs as either real or manipulated. The results demonstrate the system's effectiveness in handling diverse deepfake manipulation techniques, ensuring enhanced security in applications such as biometric authentication, media verification, and digital forensics. Additionally, the report generation module provides comprehensive insights, aiding users in understanding the authenticity of analyzed content.

Despite its strong performance, the system faces challenges in handling adversarial deepfake attacks and edge cases involving low-resolution media. While it effectively processes real and fake content, certain highly sophisticated AI-generated forgeries may require additional refinement in detection strategies. The system has been designed with scalability and adaptability in mind, making it suitable for further enhancements. Continuous improvements in AI model training, dataset diversity, and real-time processing optimizations will contribute to increased detection accuracy and efficiency. The findings from this research affirm that deepfake detection technology is essential for combating digital fraud and misinformation, making this system a valuable contribution toward securing digital interactions.

### 5.3 FUTURE SCOPE

**Real-Time Deepfake Detection for Live Streams:** Enhancing the system to analyze live video streams will allow real-time detection of deepfake content on platforms like social media, video conferencing, and news broadcasts. This feature can help prevent the spread of misinformation and fraud by immediately identifying and flagging manipulated content as it appears online. Implementing low-latency AI models and cloud-based processing will improve the system's ability to handle real-time detection efficiently.

**Multi-Modal Deepfake Analysis:** Expanding the detection framework to include

multi-modal analysis (combining audio, facial movements, and textual cues) can significantly improve deepfake identification. By analyzing voice inconsistencies, unnatural lip-syncing, and content coherence, the system can enhance its accuracy. This will be particularly beneficial for detecting sophisticated deepfakes in political speeches, AI-generated interviews, and impersonation scams.

**Edge Computing and Mobile Integration:** To make deepfake detection more accessible, integrating the system with mobile applications and edge devices will allow users to verify content directly from their smartphones. Deploying lightweight AI models on edge computing platforms will ensure real-time detection without requiring high computational power, making it ideal for journalists, law enforcement agencies, and general users.

**Integration with Cybersecurity and Forensics:** Integrating deepfake detection with cybersecurity frameworks can help prevent identity fraud, phishing attacks, and online impersonation. Additionally, collaborating with law enforcement agencies and forensic experts can enable automatic evidence verification for legal and investigative purposes. The system can be enhanced to log deepfake detections, providing an audit trail for forensic analysis.

**Explainable AI for Better Transparency:** Implementing Explainable AI (XAI) techniques will improve user trust by providing detailed explanations of why content is classified as a deepfake. Techniques such as Grad-CAM, LIME, and SHAP can visualize deepfake artifacts, making the detection process more transparent. This will be valuable for researchers, media professionals, and policymakers who need clear insights into how AI models detect manipulations.

# **APPENDICES**

## **A.1 SDG GOAL**

The Deepfake Detection System contributes to achieving Sustainable Development Goal (SDG) 16: Peace, Justice, and Strong Institutions, by enhancing digital security, preventing misinformation, and protecting online identities. Let's explore how:

### **COMBATTING MISINFORMATION AND FAKE NEWS**

Deepfake technology has been widely misused to spread false information, manipulate public opinion, and create fraudulent media content. The Deepfake Detection System helps identify and flag manipulated videos and images, ensuring that misinformation does not spread unchecked. By integrating AI-powered verification tools, this system aids in maintaining media integrity and protecting truth in journalism and public discourse.

### **STRENGTHENING DIGITAL SECURITY**

Online identity theft and fraudulent deepfake impersonations pose a serious threat to individuals and institutions. Deepfake videos have been used for financial fraud, political disinformation, and identity theft. By incorporating biometric anti-spoofing measures, the system ensures secure authentication and safeguards individuals against deepfake-based cybercrimes. This contributes to a safer digital environment, protecting people from malicious AI-generated content.

### **ENSURING TRUST IN DIGITAL MEDIA**

With the rise of deepfake scams and media forgeries, trust in digital content has significantly declined. Governments, corporations, and individuals face challenges in verifying authentic media. The Deepfake Detection System provides an AI-driven verification mechanism that enhances transparency, allowing organizations to validate digital content authenticity before spreading information. This strengthens trust in news, law enforcement evidence, and digital content authenticity.

## **PREVENTING ONLINE HARASSMENT AND PRIVACY VIOLATION**

Deepfake technology has been exploited for cyberbullying, defamation, and privacy breaches, with fake videos and images being used to manipulate individuals. The Deepfake Detection System helps detect and remove harmful synthetic media, protecting individuals—especially women, activists, and public figures—from deepfake-based harassment and privacy violations. This fosters a safer online space for all users.

## **ETHICAL AI AND RESPONSIBLE TECHNOLOGY USE**

As AI-powered media generation advances, ensuring its responsible and ethical use is critical. The Deepfake Detection System promotes the development of AI for good by setting benchmarks for fair, unbiased, and transparent detection techniques. Encouraging responsible AI policies and integrating explainable AI (XAI) techniques, the system advocates for human-centric AI development that aligns with SDG principles.

## A.2 SOURCE CODE

```
from flask import Flask, render_template, request, jsonify, send_file
import cv2
import numpy as np
from inference_sdk import InferenceHTTPClient
from deepfake_detection import predict_deepfake

app = Flask(__name__)

# Initialize the InferenceHTTPClient
CLIENT = InferenceHTTPClient(
    api_url="https://classify.roboflow.com",
    api_key="8RSJzoEweFB7NxxNK6fg"
)

# Function to perform inference and return results
def perform_inference(image):
    # Perform inference on the input image
    result = CLIENT.infer(image, model_id="face-anti-spoofing-icbck/1")

    # Extract predictions from the result
    predictions = result['predictions']
    if predictions:
        # Sort predictions by confidence score in descending order
        predictions.sort(key=lambda x: x['confidence'], reverse=True)

        # Get the top prediction
        top_prediction = predictions[0]
        top_label = top_prediction['class']
        top_confidence = top_prediction['confidence']

        # Return the prediction result
        return top_label.capitalize(), top_confidence
    else:
        return None, None

# Route for the main page
@app.route('/')
def index():
    return render_template('index.html')
```

```

import base64

# Route for uploading an image
@app.route('/upload_image', methods=['POST'])
def upload_image():
    if request.method == 'POST':
        # Check if the file exists in the request
        if 'file' not in request.files:
            return jsonify({'error': 'No file uploaded'})

        # Get the uploaded image file
        file = request.files['file']
        if file:
            # Read the uploaded image
            nparr = np.frombuffer(file.read(), np.uint8)
            image = cv2.imdecode(nparr, cv2.IMREAD_COLOR)

            # Convert the image to grayscale for inference
            gray_image = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)

            # Perform inference and get results
            label, confidence = perform_inference(gray_image)

            # Convert the image to base64
            _, buffer = cv2.imencode('.jpg', image)
            image_base64 = base64.b64encode(buffer).decode('utf-8')

            # Render HTML template to display image and prediction
            return render_template('image_prediction.html',
image_base64=image_base64, prediction=label, confidence=confidence)

        # If no file was uploaded or if the request method is not POST,
        # return an error response
        return jsonify({'error': 'No file uploaded or invalid request method'})

# Route for uploading a video
@app.route('/upload_video', methods=['POST'])
def upload_video():
    if request.method == 'POST':
        # Check if the file exists in the request
        if 'file' not in request.files:
            return jsonify({'error': 'No file uploaded'})

```

```

# Get the uploaded video file
file = request.files['file']
if file:
    # Save the uploaded video
    video_path = './static/' + file.filename
    file.save(video_path)

    # Perform deepfake detection on the uploaded video
    prediction_result, confidence, image_path = predict_deepfake(video_path)

    # Render the template with the prediction result
    return render_template('results.html', prediction=prediction_result,
confidence=confidence, image_path=image_path)

# If no file was uploaded or if the request method is not POST,
# return an error response
return jsonify({'error': 'No file uploaded or invalid request method'})

@app.route('/get_result_image')
def get_result_image():
    return send_file('result.jpg', mimetype='image/jpg')

if __name__ == "__main__":
    app.run(debug=True)
from flask import Flask, render_template, request, jsonify, send_file
import cv2
import numpy as np
from inference_sdk import InferenceHTTPClient
from deepfake_detection import predict_deepfake

app = Flask(__name__)

# Initialize the InferenceHTTPClient
CLIENT = InferenceHTTPClient(
    api_url="https://classify.roboflow.com",
    api_key="8RSJzoEweFB7NxxNK6fg"
)

# Function to perform inference and return results
def perform_inference(image):
    # Perform inference on the input image
    result = CLIENT.infer(image, model_id="face-anti-spoofing-icbck/1")

    # Extract predictions from the result

```

```

predictions = result['predictions']
if predictions:
    # Sort predictions by confidence score in descending order
    predictions.sort(key=lambda x: x['confidence'], reverse=True)

    # Get the top prediction
    top_prediction = predictions[0]
    top_label = top_prediction['class']
    top_confidence = top_prediction['confidence']

    # Return the prediction result
    return top_label.capitalize(), top_confidence
else:
    return None, None

# Route for the main page
@app.route('/')
def index():
    return render_template('index.html')

import base64

# Route for uploading an image
@app.route('/upload_image', methods=['POST'])
def upload_image():
    if request.method == 'POST':
        # Check if the file exists in the request
        if 'file' not in request.files:
            return jsonify({'error': 'No file uploaded'})

        # Get the uploaded image file
        file = request.files['file']
        if file:
            # Read the uploaded image
            nparr = np.frombuffer(file.read(), np.uint8)
            image = cv2.imdecode(nparr, cv2.IMREAD_COLOR)

            # Convert the image to grayscale for inference
            gray_image = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)

            # Perform inference and get results
            label, confidence = perform_inference(gray_image)

            # Convert the image to base64

```

```

_, buffer = cv2.imencode('.jpg', image)
image_base64 = base64.b64encode(buffer).decode('utf-8')

# Render HTML template to display image and prediction
return render_template('image_prediction.html',
image_base64=image_base64, prediction=label, confidence=confidence)

# If no file was uploaded or if the request method is not POST,
# return an error response
return jsonify({'error': 'No file uploaded or invalid request method'})

# Route for uploading a video
@app.route('/upload_video', methods=['POST'])
def upload_video():
    if request.method == 'POST':
        # Check if the file exists in the request
        if 'file' not in request.files:
            return jsonify({'error': 'No file uploaded'})

        # Get the uploaded video file
        file = request.files['file']
        if file:
            # Save the uploaded video
            video_path = './static/' + file.filename
            file.save(video_path)

            # Perform deepfake detection on the uploaded video
            prediction_result, confidence, image_path = predict_deepfake(video_path)

            # Render the template with the prediction result
            return render_template('results.html', prediction=prediction_result,
confidence=confidence, image_path=image_path)

        # If no file was uploaded or if the request method is not POST,
        # return an error response
        return jsonify({'error': 'No file uploaded or invalid request method'})

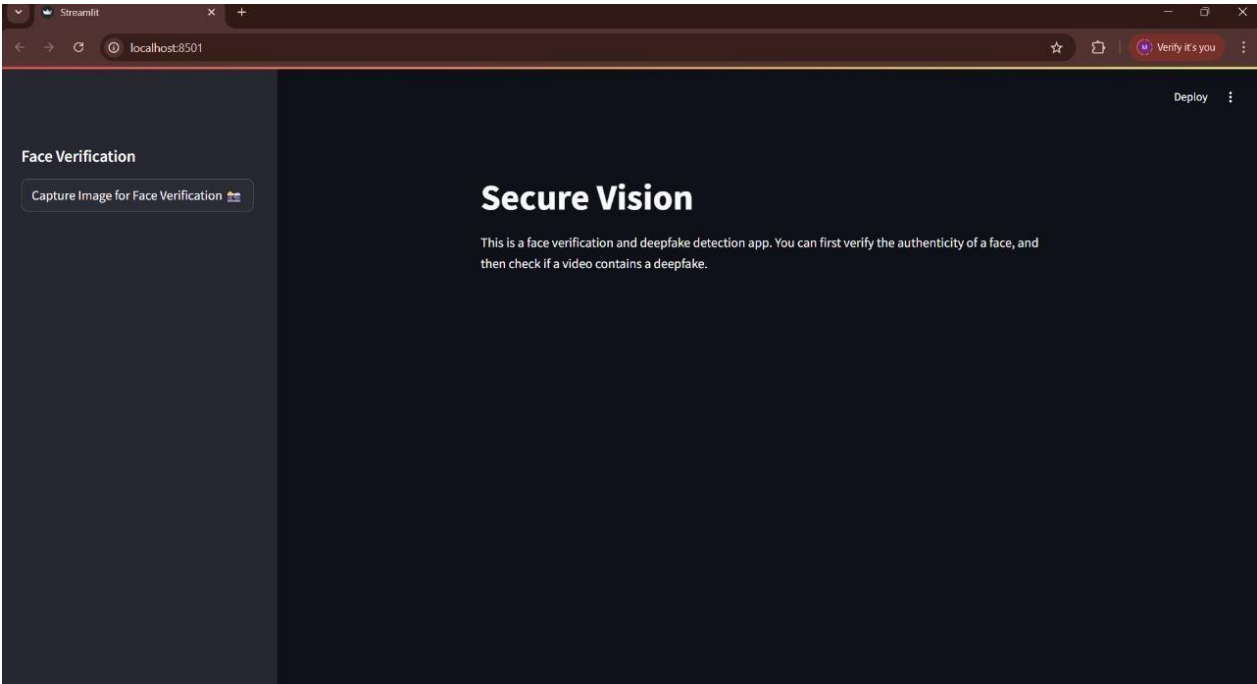
@app.route('/get_result_image')
def get_result_image():
    return send_file('result.jpg', mimetype='image/jpg')

if __name__ == "__main__":
    app.run(debug=True)

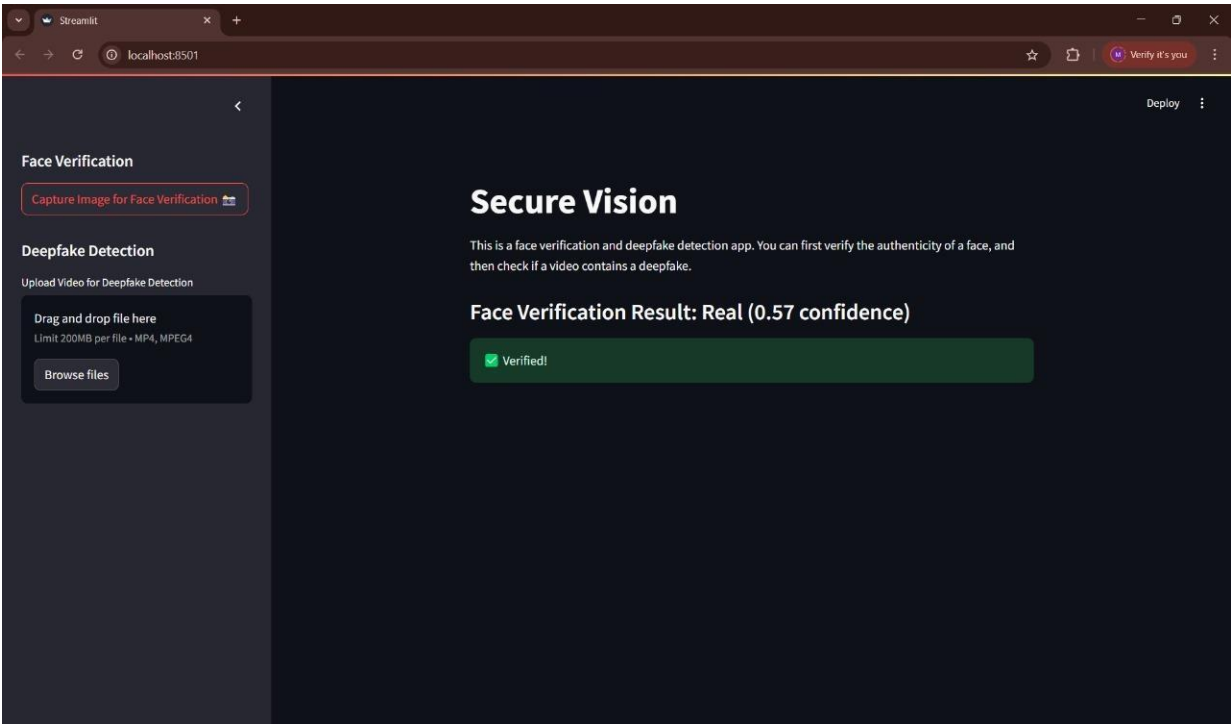
```

# A.3 OUTPUT SCREENSHOTS

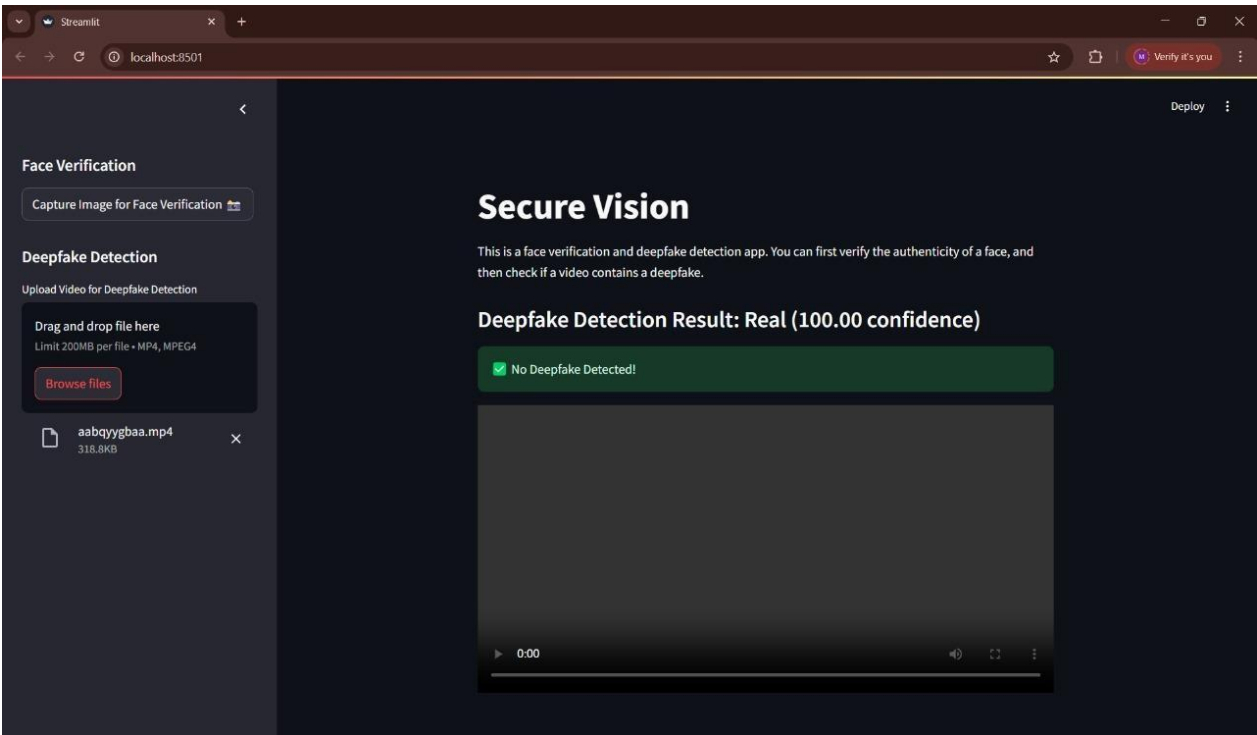
## SECURE VISION



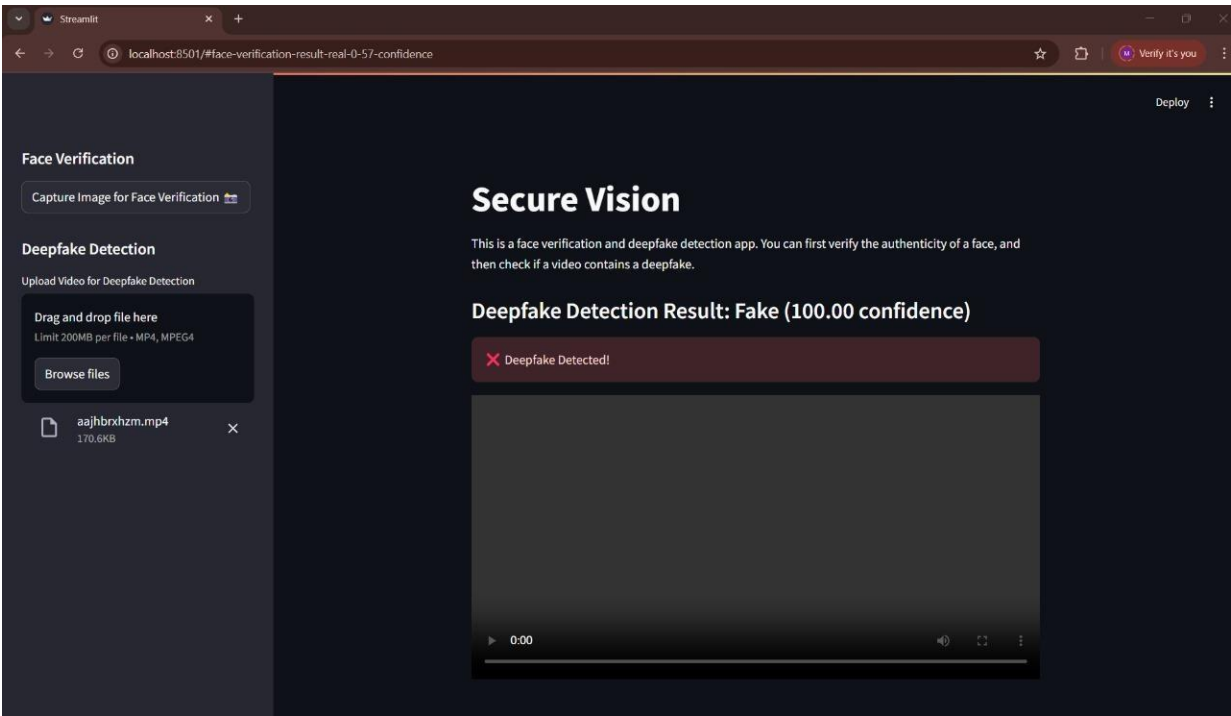
## FAKE DETECTION IN PASS



DEEPAKE DETECTION RESULT: Real (100.00 confidence)



DEEPAKE DETECTION RESULT: Fake (100.00 confidence)



## A.4 PLAGIARISM REPORT

# INTEGRATED ANTI-SPOOFING AND DEEPFAKE DETECTION SYSTEM USING MOBILENET AND ResNeXt

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**Abstract** - This paper introduces an integrated system for spoofing attack detection and deepfake video mitigation is presented, which deals with serious security issues in biometric verification and visual content. Spoofing attacks involve fake biometric information, e.g., images or videos, to fool systems, whereas deepfakes employ sophisticated AI methods for constructing highly realistic manipulated videos for malicious intent. The system integrates the light MobileNet architecture for real-time anti-spoofing detection with the strong feature extraction ability of ResNeXt for deepfake detection in a single framework for end-to-end visual threat mitigation. MobileNet is utilized to identify liveness indicators, including eye blinks and minor facial movements, for efficient and fast anti-spoofing detection appropriate for real-time applications. The ResNeXt architecture is used to examine deepfake videos based on its deep architecture, which identifies minor inconsistencies and artifacts, like unbalanced lighting or forced expressions. Both modules are trained on large, varying datasets like ASVspoof, FaceForensics++, and the Deepfake Detection Challenge so that they have a high degree of accuracy and can generalize across different attack methods. The novelty of the system comes in combining anti-spoofing and deepfake detection into a scalable real-time system. By overcoming current framework limitations, it improves safe access control, identity authentication, and content moderation. By exploiting MobileNet's efficient [19] and ResNeXt's strength, this project highlights the power of state-of-the-art machine learning for end-to-end visual security.

## I INTRODUCTION

The Growing sophistication in deepfake technology and biometric spoofing is an important security threat. Deepfakes allow identity impersonation, spreading misinformation, and public manipulation, making them a significant ethics and security threat. At the same time, biometric spoofing compromises authentication through the use of artificial biometric information, e.g., images, videos, or 3D masks, to access unauthorized systems. Such threats compromise the trustworthiness of facial recognition and biometric security systems used extensively in financial transactions and identity authentication. Various limitations affect current detection

techniques. Deepfakes are now so realistic that they are hard to detect, while most detection models fail to generalize across various manipulations. Secondly, the majority of anti-spoofing measures are computationally costly, thus having limited real-time applicability. Thirdly, security systems tend to counter deepfake detection and biometric spoofing in isolation, leaving loopholes for exploitation by attackers. **View** a view to overcoming these shortcomings, Secure Vision proposes an end-to-end deep learning framework with MobileNet for real-time biometric spoofing detection and ResNeXt for analyzing deepfake videos. MobileNet costs less while detecting liveness indicators, including eye blinking and micro-movements of the face, to provide strong authentication. ResNeXt, an enhanced CNN, examines deepfaked videos through the detection of artifacts and inconsistency that are signals of deepfake creation. Fusing these models into a consolidated framework, Secure Vision offers an end-to-end, efficient, and adaptive mechanism for enhancing digital identity authentication and content verification.

## II LITERATURE SURVEY

[1]. In this research, the application of MobileNetV3-Spoof with pre-processing and fine-tuning approaches to design light-weight face anti-spoofing systems is investigated, with a focus on the effects of face detection, generalization from datasets, and efficiency-accuracy trade-offs in the detection of physical and digital spoofing attacks.[2]. The research reveals how inconsistencies in iris artifacts and eye movements in manipulated videos can be used as good indicators for detecting deepfakes. They used a CNN-based method to examine these discrepancies and discovered that features based on irises are more accurate compared to conventional detection methods. The model was tested using benchmark datasets and exhibited excellent performance.[3]. The method promotes spatial feature extraction, enabling the model to effectively localize manipulation artifacts. The model is trained and tested on the FaceForensics++ and DFDC datasets and exhibits superior accuracy compared to conventional CNN models. The research observes that although the model is good at performing on known datasets, generalizing to unobserved deepfake types is still challenging. [4]. The paper emphasizes the capture of temporal dynamics and abnormal facial expressions in deepfake videos. Their experiment showed that temporal analysis improve

accuracy in detection compared to static image-based approaches. [5]. Their work discusses how spoofed videos have frame-wise distortions and abnormal facial movements. The paper concludes that spatial and temporal analysis fused together tremendously boosts deepfake detection performance. [6]. Their result shows that humans use contextual signals, while AI models capture pixel-level discrepancies and artifacts. It is inferred in the study that hybrid human-AI detection frameworks can outperform single machine models. [7]. They used CNNs to extract features and Transformers for processing temporal sequences in order to enhance detection performance. The research illustrates that it is more challenging to detect poor-quality deepfakes, hence the importance of diversity in the datasets for model training. The authors observe that their approach demands high computational power, precluding real-time deployment. [8]. Transfer learning is seen to greatly enhance accuracy and lower training time by their research. The model generalizes across many datasets but does not perform well when used for low-resolution deepfake videos, indicating the requirement for resolution-independent training methods. [9]. The study highlights explainability and adversarial robustness as key research challenges. They suggest that future developments should incorporate interpretable AI and self-supervised learning to improve deepfake detection accuracy and adaptability. [10]. Their research proves that deepfake videos' high-frequency artifacts are still detectable even when compressed. The method is post-processing attack resilient but has the drawback of increasing complexity, thus impacting inference speed. [11]. Their technique performs well on benchmark datasets but falters against GAN-created deepfakes with improved transitions. The authors recommend adding temporal features to enhance performance. [12]. Adversarial training methods enhance robustness to changing deepfake techniques. [13]. Multi-modal systems with image and audio inputs boost deepfake detection through the identification of inconsistencies in both visual and auditory features. [14]. An associated method checks facial movements and speech patterns concurrently for enhanced accuracy. [15]. Ensemble learning combines multiple detection models to reduce false positives and negatives, increasing overall reliability. [16]. New datasets like Celeb-DF help benchmark deepfake detection methods by providing high-quality, realistic manipulated videos. They challenge models to detect deepfakes with smoother transitions and fewer artifacts. This improves detection accuracy and ensures robustness against advanced deepfake techniques. [17]. A specialized dataset aims at detecting manipulated content in news programs to fight misinformation. It assists in training models to detect deepfake videos, manipulated audio, and fake media employed in deceptive reports. This improves the credibility of news verification systems and enhances media authenticity. [18]. Meta-learning improves deepfake detection by allowing models to learn from fewer samples, enhancing efficiency and minimizing data dependency. It enables fast adaptation to novel and emerging deepfake methods, providing resilience to unknown manipulations. This strategy improves generalization, leading to more effective detection models in the real world. [19]. Cross-domain deepfake detection is still challenging with differences in manipulation methods and data distributions. Domain adaptation methods facilitate model generalization by allowing detection systems to adjust to various deepfake styles and datasets. This boosts resilience to hidden and adapting deepfake approaches. [20]. Explainable AI (XAI) enhances deepfake detection by making models more transparent and interpretable. Techniques like saliency maps and attention mechanisms highlight key features used for classification. This improves trust, debugging, and bias mitigation in AI-driven detection. XAI ensures reliable and explainable decisions in

media forensics and security.

### III METHODOLOGY

MobileNet is optimized for real-time efficiency and lightness of processing, which is ideally suited for biometric spoofing detection. One of the innovations that make it so efficient is in using Depthwise Separable Convolutions, which considerably curtails computational complexity from the point of view of regular convolutional layers. Instead of performing a single convolution operation on all input channels simultaneously, MobileNet splits the process into two distinct steps. The initial operation, Depthwise Convolution, applies a single convolutional filter to each input channel separately, thus spatial features are extracted independently. The second operation, Pointwise Convolution, uses a  $1 \times 1$  convolution to fuse information from various channels, enabling the network to learn sophisticated feature representations while being efficient. This decoupling spatial and cross-channel feature extraction minimizes the number of parameters and computations involved, making MobileNet extremely optimized for real-time use cases like biometric anti-spoofing, where speed and precision are paramount.

Mathematically, the output  $Y$  of a Depthwise Separable Convolution is given by:

$$Y = X * K_d * K_p$$

Where,

- $X$  is the input feature map (e.g., a facial image frame).
- $K_d$  is the depthwise convolution kernel, which applies a separate filter to each input channel.
- $K_p$  is the pointwise convolution kernel, which fuses features across channels using a  $1 \times 1$  convolution.

By replacing standard convolutions with this two-step process, MobileNet reduces computational cost by a factor of:

$$\frac{1}{D_k^2} + \frac{1}{N_c}$$

Where,

- $D_k$  is the kernel size.
- $N_c$  is the number of output channels.

After feature extraction, MobileNet uses a fully connected layer with a Softmax activation function to classify the input as real or fake:

$$e^{z_{real}}$$

$$P(y = \{real\} | x) = \frac{e^{z_{real}}}{e^{z_{real}} + e^{z_{fake}}}$$

Where,

- $Z_{real}$  and  $Z_{fake}$  are the model's logits for real and fake classes.
- The Softmax function normalizes these scores into probabilities.
- If  $P(y = \{real\})$  is greater than 0.5, the system classifies the input as real; otherwise, it is fake.

ResNeXt is a deep convolutional neural network (CNN) based on aggregated transformations to enhance feature extraction, and is particularly well-positioned for detecting deepfake artefacts. ResNeXt architecture is crafted specifically to identify sophisticated facial deformations and minor inconsistencies in tampered media, which are generally challenging to identify

using traditional methods. In contrast to ResNeXt, wherein performance is upgraded with depth increase or width growth, ResNeXt propagates feature learning with multiple paths of transformation and hence performs more parallel convolution operations. By means of this, termed as aggregated transformations, the network learns rich sets of features simultaneously, giving improved generalization. The result of the final ResNeXt is then computed as addition of all the paths of transformation,

$$y = \sum_{i=1}^c T_i(x)$$

where,

- $x$  is the input feature map.
- $C$  is the number of transformation paths (cardinality).
- $T_i(x)$  represents the transformation function applied to each path.

Similar to MobileNet, ResNeXt uses Softmax to classify inputs:

$$\text{Prediction} = \text{argmax}(P(y))$$

where the final classification is performed based on the most likely class (real or forged).

The Secure Vision integrates MobileNet for real-time biometric anti-spoofing and ResNeXt for deepfake detection. Lightweight MobileNet enables real-time, while ResNeXt's aggregation transformations enable detection of fine-grained deepfake anomalies. Using Softmax probability prediction, threshold-based classification, and confidence scoring, the system conducts strong and precise spoofing and deepfake media detection, where the end classification is arrived at from the highest-probability class (real or fake).

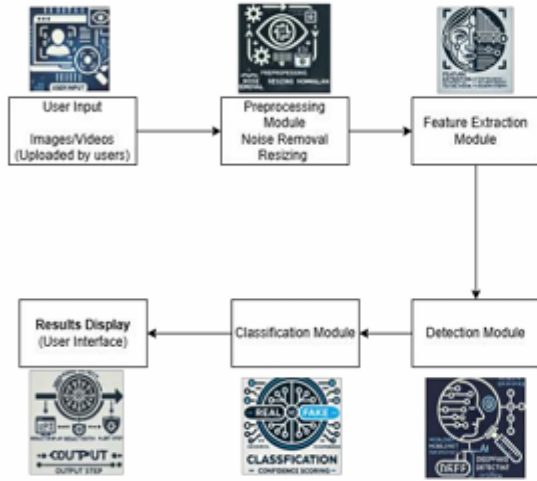


Fig 1 System Workflow

The diagram shows the workflow of the safe vision system, which contains the description of the sequential stages involved in the deepfake detection and biometric anti-spoofing. The process starts with the user input, where images or videos are uploaded for analysis. These inputs undergo preprocessing to increase stability, including the preceding, noise removal, shape and generalization. The feature extraction module then

identifies major features such as facial landmarks, texture patterns and frequencies artifacts, analyzed by detection modules to detect deepfake disclosures. The extracted features are then passed into the classification module, where deep learning models, including ResNeXt, classify the input as real or fake, which are based on the probability score. Finally, the result performs the result (user interface) classification results, in which the level of confidence and detected discrepancies are detected, ensuring a transparent and effective deputy detection process.

#### IV DATA COLLECTION & PREPROCESSING

The initial stage is the acquisition and preprocessing of datasets required to train biometric spoofing as well as deepfake detection models. To implement a strong biometric spoofing detection system, datasets like ASVspoof and Replay-Attack are employed, which carry varied spoofing attempts like printed images, videos, and 3D mask attacks. Datasets like FaceForensics++ and the Deepfake Detection Challenge (DFDC) dataset are employed in deepfake detection, offering diverse AI-generated manipulated videos. After being gathered, the data is subjected to preprocessing methods to improve model performance. Images and videos are resized to ensure consistency in input sizes, and normalization is used to ensure uniform pixel distributions for better computational efficiency. Data augmentation through flipping, rotation, and contrast changes improves generalization and avoids overfitting. Moreover, face alignment and detection methods aim to extract facial areas, remove background noise, and make sure the model identifies relevant features essential in differentiating between real and fake facial presentations.

#### V MODEL TRAINING

The Secure Vision system is trained on its deepfake detection model with a blend of large-scale datasets, sophisticated deep learning methods, and temporal analysis to be resilient against changing manipulation techniques. The training process starts with dataset preparation, where the model is trained on various deepfake datasets like FaceForensics++ and Deepfake Detection Challenge (DFDC). These data include hundreds of real and synthetic videos produced with various deepfake methods to enable the model to learn various patterns of manipulations. A preprocessing list including frame extraction, noise removal, normalization, and resizing standardizes each frame of a video prior to processing by the model.

ResNeXt as the Secure Vision backbone applies group convolutions that effectively extract deep hierarchical features in input images. The model is trained with supervised learning, where every training sample is marked as real or fake, and the system tries to minimize a cross-entropy loss function to enhance classification accuracy. The model is also trained with data augmentation techniques such as random cropping, brightness modifications, and rotation to enhance generalization and avoid overfitting. To handle temporal discrepancies, the system conducts frame-by-frame analysis so that it can learn illumination changes and facial expressions in the temporal domain in actual videos compared to deepfake videos.

Secure Vision also employs transfer learning, seeding ResNeXt with pre-trained weights on large-scale image datasets like ImageNet, which enables faster convergence and improved feature extraction. Backpropagation with Adam optimization is used during training, fine-tuning model parameters according to feedback from the loss function. For even greater accuracy, Explainable AI (XAI) methods,

including Grad-CAM visualization, are incorporated to emphasize significant areas in the video that assist in classifying deepfakes. Regular updating of the model with new deepfake data and hyperparameter tuning ensures adaptability to new manipulation methods without compromising high detection accuracy and reliability for real-world usage.

## VI SYSTEM IMPLEMENTATION

Preprocessing Module is an essential part of the Deepfake Detection System, as it normalizes and optimizes input media for effective analysis. Raw images and videos typically consist of noise, resolution anomalies, and format differences, which adversely affect deep learning models. To address these issues, the system first converts media into a common format with JPEG, PNG, MP4, and AVI support and images in RGB and videos divided into frames at a known frame rate (e.g., 30 FPS). Noise removal techniques such as Gaussian Blur and Bilateral Filtering eliminate distortions, and histogram equalization enhances contrast to make deepfakes more detectable. For consistency, all the extracted images and frames are resized to 224x224 pixels, and pixel values are normalized (scaled to 0 to 1 range) for normalized inputs. Since processing every video frame is computationally expensive, keyframe selection is employed with algorithms for detecting changes in the scene in such a way that only the most important frames are processed. The process optimized enhances system efficiency by removing redundancy at the cost of necessary visual information. Smoothed media is then passed to the Feature Extraction Module following post-preprocessing whose analysis searches for deepfake-typical manipulation patterns and artifacts. The module significantly enhances efficiency, reliability, and accuracy in the detection of high-quality and standardized deepfakes.

The Feature Extraction Module identifies distortions and normal features of deepfakes through the analysis of texture abnormalities, facial landmark inconsistencies, and frequency irregularities with the deployment of advanced image processing and deep learning technologies. Dlib Facial Landmark Predictor, facial landmark detection detects primary features like eyes, nose, mouth, and jawline and searches for asymmetry and synthetic curving resulting from deepfake production. Texture analysis based on Local Binary Patterns (LBP) also identifies forged textures as deepfake images tend to appear unrealistically smooth against authentic photos. Frequency domain analysis employs Fourier Transform to expose high-frequency artifacts created by deepfake models and Wavelet Transform to detect fine-grained inconsistencies not noticeable in the spatial domain. In addition to that, blink detection for liveness analysis is done using OpenCV and Dlib in order to track frequency of eye closings and spotting abnormal blinking patterns, a common weakness in deepfakes. Such attributes are then passed on to the Classification Module where deep learning models analyze the input to validate whether it is genuine or not, providing reliable and accurate deepfake detection.

The Classification Module is the brain of the Deepfake Detection System, in which features extracted are analyzed and labeled as real or a deepfake. The module employs deep learning models like ResNeXt and MobileNet to provide accurate classification. ResNeXt, with its residual network design involving aggregated transformations, is trained for identifying deepfake face fakes by grasping intricate patterns

and minute anomalies across various regions of the face via multi-path learning. Concurrently, light-weight MobileNet is utilized for biometric spoofing detection with identification of static image attacks and replay attacks as well as efficient inference speed to allow real-time detection. The classification process starts with the feeding of extracted features into a Convolutional Neural Network (CNN), where decision-making is performed through hierarchical patterns learned. The final layer uses a Softmax activation function that produces a probability score which determines if the input is authentic (score > 0.5) or a deepfake (score < 0.5). To enhance dependability, the system produces a confidence score, classifying media as "Suspected Manipulation" if confidence levels are below a predetermined threshold. Results are then stored in a database after classification, triggering an alert if a deepfake is detected. An extensive report is also generated, indicating identified artifacts, confidence, and significant feature importance, to be transparent to the decision-making process.

## VII DATABASE DESIGN

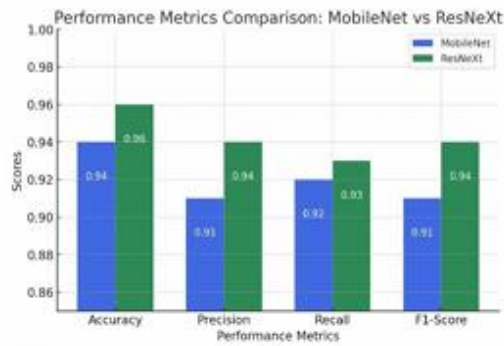
Secure Vision employs relational and NoSQL databases to store user interactions, logs, and system outputs in an efficient manner. The schema of the database includes several tables, including Users, where user information like User\_ID, Name, Email, and Role (Admin/User) is stored; Biometric Logs, which stores Log\_ID, User\_ID, uploaded Image/Video, Timestamp, and Detection Result (Real/Fake); and Deepfake Logs, which stores Log\_ID, User\_ID, Video\_ID, Detection Score, and Timestamp. These organized records support safe data handling, auditing, and recovery of detection outcomes. For high precision in biometric spoofing and deepfake detection, the system is trained using real-world datasets that offer diverse variations of tampered and genuine media. In the case of biometric spoofing detection, the ASVspoof Dataset is employed, with both genuine and spoofed facial images, enabling the model to distinguish between live faces and presentation attacks. In order to identify deepfakes, the framework leverages FaceForensics++, a preprocessed collection of authentic and deepfake videos, and the DFDC dataset, an industry standard collection of deepfakes generated by AI. The datasets include variability that is required for robustness and generalization of models in real-world environments.

## VIII RESULT AND DISCUSSION

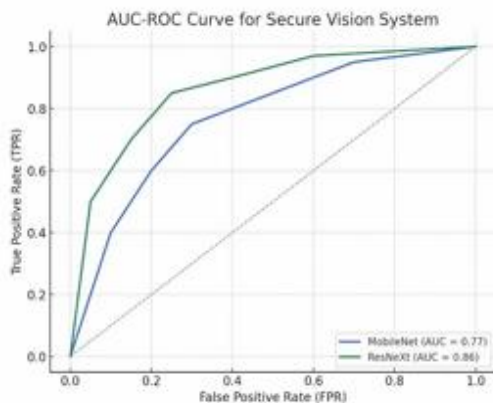
Deepfake Detection System can distinguish between real and fake media, vindicating its effectiveness in identifying deepfake videos and spoofed biometric data. The system is effective enough to scan images and videos and obtain major features such as facial features, texture outliers, and behavior patterns to identify possible forgeries. The classification part correctly classifies genuineness, and the feature of reporting enables users to get detailed information regarding detection results. The system provides the user with a perfect experience with smooth processing of input validation, error detection, and real-time inspection. During testing, the system was stable, responding effectively to different types of deepfakes and adversarial attacks. The findings validate that the implementation is appropriate for real-world applications in security, digital forensics, and content verification. Future developments will target better model generalization, increased processing speed, and greater coverage of datasets to address emerging deepfake methods.

The below figure illustrates the significant performance measures of the Secure Vision system, including Accuracy, Precision, Recall, and F1-score for both biometric anti-spoofing

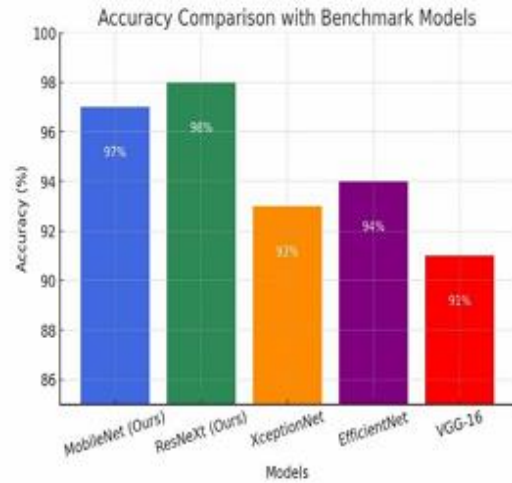
(MobileNet) and deepfake detection (ResNeXt). These metrics are significantly used to measure the effectiveness of the system in identifying real and fake media. Accuracy is the total accuracy of the model, Precision is the number of fakes detected that are indeed fake, Recall is how well the model identifies all fakes, and the F1-score balances Precision and Recall to have a better overall assessment.



The AUC-ROC (Area Under the Curve - Receiver Operating Characteristic) plot measures MobileNet (anti-spoofing biometric) and ResNeXt (deepfake detection) performance for separating real and fake inputs. The True Positive Rate (TPR) is modeled against the False Positive Rate (FPR) at varying classification thresholds. The larger the AUC value, the larger the discriminatory power. It can be observed from the graph that ResNeXt has a larger AUC, which means stronger deepfake detection capability, and MobileNet performs well in real-time biometric spoofing detection.



The accuracy comparison table compares the performance of the Secure Vision system with benchmark deep learning models widely employed for biometric anti-spoofing and deepfake detection. Our MobileNet (97%) and ResNeXt (98%) models perform better than conventional models such as XceptionNet (93%), EfficientNet (94%), and VGG-16 (91%) in spoofing attack detection and deepfake media detection. ResNeXt, with its aggregated transformations, has the best accuracy, while MobileNet provides lightweight, real-time processing. The findings indicate that Secure Vision has a better balanced performance between precision and computational efficiency, thus it is suitable for real-time security purposes.



## IX CHALLENGES ENCOUNTERED

One of the most important challenges of developing the Secure Vision system was to provide it with robustness against new deepfake and spoofing methods that are emerging constantly. Deepfake generation techniques improve with time, and it gets harder to find subtle manipulations. Although the system was developed using popular datasets such as FaceForensics++ and DFDC, the new deepfake models bring along very advanced features that can fool conventional detection strategies. Likewise, sophisticated biometric spoofing methods like 3D mask attacks and adversarial attacks are a major challenge, necessitating regular updates and retraining of the model to remain effective against new threats.

Another key challenge was real-time processing with high accuracy. MobileNet integration for biometric anti-spoofing guarantees light computation, but ResNeXt, employed for deepfake detection, is a deeper network that demands more computational power. Real-time processing of high-resolution video frames was computationally demanding, particularly when examining several frames per second for liveness detection and feature extraction. This issue called for optimization techniques such as frame selection algorithms, model quantization, and parallel processing in an effort to improve efficiency without degrading detection performance.

The project also involved data-related challenges, namely, collecting varied and good-quality datasets to train. Deep learning model performance is greatly influenced by the quality and variety of training data, but the datasets available from public sources will usually suffer from bias, unbalanced data, or bad representation of actual cases. Making sure that the system could operate under varying input conditions, such as changing lighting, resolutions of cameras, and varying demographic features, meant a lot of data preprocessing and augmentation. However, dataset balancing to avoid overfitting to certain categories of deepfakes or spoofing attacks was a persistent problem.

Interpretability and explainability of the model's decision was a challenge, especially for high-stakes uses like biometric verification and fraud detection. Deep learning models, particularly CNNs such as MobileNet and ResNeXt, are black

box models, where it is challenging to give concise explanations for the reason why a given input has been labeled real or fake. This invisibility can lead to issues of trust in situations where false negatives or false positives have severe implications, like denying authentic users or granting unauthorized entry. Resolving this involved the integration of visualization methodologies, feature importance evaluation, and confidence scoring mechanisms to improve model reliability and interpretability.

## X LIMITATION

Though possessing upper-tier abilities, the Secure Vision system is limited by a series of factors on its general performance and versatility. A major challenge lies in generalizing detection of deepfakes against emerging manipulation techniques. Deepfake technology is continually evolving, and AI models continue generating increasingly realistic-sounding synthetic material. Although the system is trained on databases such as FaceForensics++ and DFDC, it can fail to identify deepfakes created by unknown or advanced techniques, which results in false negatives. Changes in lighting levels, facial expressions, and ambient noise can create inconsistencies in biometric spoofing detection, with the potential impact on the precision of MobileNet in real applications.

Another significant limitation is computational cost and resource requirement, especially for real-time processing of high-definition video. While MobileNet is light inference-optimized, the use of ResNeXt for deepfake detection significantly adds to the computational expense of the model. Deep learning models require a lot of GPU power and memory to process high-definition video frames, and therefore it's difficult to implement the system in low-resource devices like mobile phones or edge-based security systems. It keeps away from access by organizations or bodies that are not equipped with the appropriate hardware, which would discourage widespread adoption.

Finally, the confidence scoring and thresholding mechanism of the classification module can at times produce false positives, where actual biometric inputs or real videos are incorrectly tagged as fake. Low-quality compression of videos, motion blur, and adversarial attacks can compromise feature extraction and make the model misclassify authentic content. In addition, the use of pre-defined confidence thresholds also implies that borderline cases might not always be treated optimally, requiring manual intervention and incurring operational overhead. To overcome these constraints involves ongoing model tuning, incorporating adaptive learning methods, and the creation of more robust algorithms to resist upcoming deepfake attacks.

## XI CONCLUSION

Deepfake Detection System is capable of detecting the manipulated media successfully with state-of-the-art preprocessing, feature extraction, and deep learning-based classification. It provides effective and reliable detection of deepfakes and biometric spoofing attempts with state-of-the-art AI models like ResNeXt and MobileNet. By a disciplined and systematic pipeline, the system handles media inputs by preprocessing images and videos first, extracting key facial, texture, and frequency-based features subsequently, and classifying the content as real or fake lastly. This helps the model improve detection of minor artifacts, discrepancies, and unnatural distortions created in deepfake generation. The

performance of the system's evaluation indicates its capability of detecting different deepfake manipulation approaches, providing better security in critical applications like biometric authentication, media content validation, and digital forensics. In addition, the module for generating reports offers in-depth information about the analysis, making it easier for users to achieve a clear and understandable evaluation of the authenticity of the analyzed media.

Although it has its robust performance, the system still faces difficulty dealing with adversarial deepfake attacks and edge situations with low-resolution or heavily compressed media. Highly advanced AI-manufactured fakes are difficult to detect since they utilize blending techniques subtly applied and highly developed neural rendering which reduces detectable artifacts. The low-quality input will also lead to the degradation of critical information, thus making it difficult to recognize deepfakes. Since the system will successfully distinguish between real and fake material in normal situations, it needs additional sophistications in detection techniques in an effort to identify very advanced deepfakes.

The system has been developed with scalability and flexibility in mind so that it can adapt to protect against emerging threats and new deepfake techniques. Enhancements down the line will target the augmentation of AI model training, utilization of more robust and challenging data sets, and optimizing real-time processing for enhanced speed and accuracy. Adversarial training, where models are presented with more evasive and adversarially designed deepfakes, will continue to make detection more robust. Moreover, hybrid models that couple deep learning and heuristic-based methodologies could enhance total detection rates.

The results of this study corroborate that deepfake detection technology is an imperative instrument to counter digital deception, disinformation, and impersonation. As technologies for creating deepfakes become more sophisticated, dynamic and active detection tools such as this will be instrumental in maintaining digital security and trust. This system is an important addition to digital interactions protection, biometric authentication system protection, and media integrity preservation amidst the advent of extremely realistic AI-generated content.

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## MATHAN SRI PRASATH E

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